

AMBO UNIVERSITY

HACHALU HUNDESA CAMPUS

SCHOOL OF INFORMATICS AND ELECTRICAL ENGINEERING

DEPARTMENT OF INFORMATION TECHNOLOGY

OROMO TRIBE MUSIC CLASSIFICATION USING MACHINE LEARNING TECHNIQUES

By

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AMBO, ETHIOPIA

Nov, 2024 GC



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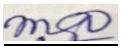
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LISTS OF ABBREVIATION

CD: compact disk

RF: Random Forest

KNN: K-Nearest Neighbors

SVM: Support Vector Machine

XGBoost: Extreme Gradient Boosting

AdaBoost: Adaptive Boosting

MFCC: Mel-Frequency Cepstral Coefficients

RMS: Root Mean Square

CM: Confusion Matrix

ABSTRACT

Music is a universal form of expression that plays a vital role in preserving cultural heritage. The objectives of study focuses on classifying audio music Oromo tribes using various machine learning techniques, as identifying different music styles can offer valuable cultural insights. However, the classification of Oromo tribal music presents significant challenges due to diverse styles, varying audio quality, and a lack of standardized classification methods, leading to inconsistencies in existing approaches. This gap highlights the need for automated tools that can accurately categorize Oromo music, facilitating better understanding and preservation of this rich heritage. To address these challenges, the specific objectives of this research includes, reviewing literature on music genre classification, finding audio features suitable for Oromo music classification, applying machine learning algorithms including classical and ensemble to classify Oromo music style and comparing and evaluating the performance of different machine learning algorithms in classifying Oromo music styles. A dataset of 2,454 audio samples, totaling 23 hours and 32 minutes which around 3.091 GB size was collected and prepared for this work. Preprocessing steps included normalization, feature extraction, and audio segmentation, utilizing features such as Mel-frequency cepstral coefficients (MFCC), root mean square (RMS), spectral features, tempo, and average beat frame had been performed. We investigated several algorithms, including Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Gradient Boosting, Extreme Gradient Boosting (XGBoost), and AdaBoost for classifying Oromo music. Our experiments, conducted with an 80% training and 20% testing split, revealed that the KNN classifier achieved the highest accuracy at 98.18%, followed by SVM at 96.56%, RF at 91.70%, XGBoost at 95%, Gradient Boosting at 95.14%, and AdaBoost at 83.20%. While the results are promising, challenges in public data unavailability and feature extraction and the need for larger datasets remain to enhance classification performance.

***Keywords**: Oromo Music, Feature Extraction, Audio Classification, Machine learning.*

CHAPTER ONE: INTRODUCTION

1.1 Background

In contemporary society, the entertainment industry significantly influences global social, political, and economic matters. In our interconnected world, countries leverage entertainment in numerous ways to advance their interests and express their cultural identities. A key aspect of this industry is the music sector, which serves as a powerful medium for sharing and promoting cultural traditions and practices on a global scale (Moges, 2023). Around the world, music serves as a deeply unifying language that bridges generations and cultures (Jagtap et al., 2024).

As noted by (Killin, 2018), music is an ancient art form that reflects numerous aspects of daily life, and every known civilization with some degree of technological advancement has embraced music as a significant cultural practice. The origins of music remain uncertain its beginnings in human history, including when and why it emerged, are still debated (Burgess, 2014). However, archaeological findings indicate that the oldest known musical instruments date back approximately 40,000 years (Killin, 2018). Many argue that the human voice is the first instrument, yet most cultures have developed their unique methods of music creation, ranging from simple percussion instruments to complex devices like the pipe organ (Cohen, 2015; Keller, 1995). In the late 1960s, the cassette tape emerged as a popular medium for music recording. Blank cassettes were sold at significantly lower prices than traditional records, despite offering the same storage capacity. The release of the Sony Walkman in 1979, along with improvements in cassette sound quality, propelled the cassette into the forefront of music consumption by 1983 (Fly, 2016). The introduction of the Compact Disc (CD) format in the late 1980s revolutionized the music industry once again. Subsequently, technology in the digital age replaced CDs, with MP3 (Motion Picture Experts Group Layer-3) becoming the dominant format, supplanting CDs and other physical media (Fly, 2016).

Audio classification entails the analysis and identification of different types of audio, including sounds, noises, musical notes, and related data, to accurately categorize them. The audio data can manifest in various forms, such as sounds produced by acoustic instruments, musical chords, human speech, or natural sounds like birdsong. Genres function as human-created labels that define or describe distinct styles of music (Jawaherlalnehru et al., 2018).

1.1.1 Significance of Music industry

Music serves as a means of communication that transcends language and cultural barriers, reflecting self-expression and identity (Lansdown & He, 2019). The music industry holds significant importance for a country from two key perspectives. Firstly, music conveys cultural identity, showcasing how a society lives, dresses, dances, and engages in daily activities through its traditional sounds. Secondly, music fosters the arts, boosts tourism, drives economic development, and creates jobs (Fly, 2016).However, effectively presenting recorded music to audiences is crucial for these benefits to materialize. As the volume of digital music produced and stored continues to rise, it becomes increasingly important to connect the right music with the appropriate audience.

1.1.2 Ethiopian Music

Ethiopia is a nation rich in diversity, home to many nations and ethnic groups, each possessing its own language, culture, music, and traditional practices. These distinct cultures are often showcased through various instruments and dance styles, both in solo and group performances (Moges, 2023). Traditional music plays a vital role in cultural and social events, including marriages, hunting, farming, gatherings, and rites of passage, thereby reinforcing community bonds.

1.1.3 Oromia and Oromo people's Music culture

Oromia, the largest regional state in Ethiopia, has a population exceeding 35 million as per the 2007 census. It spans approximately 141,699.5 square miles (367,000 km²), which accounts for over 30% of the nation's total land area and features a wide variety of geographical and ecological landscapes (Dibaba, 2013). An estimated 85% of Oromia's population resides in rural areas. The region is organized into 20 administrative districts, 8 municipal districts, 287 woredas (sub-districts), and 46 cities. According to (Jalata, 2019) ,coffee and khat are the primary cash crops cultivated in the region. Additionally, pastoral and agro-pastoral communities in Oromia rely on livestock and their products for sustenance and income.

The two primary Oromo groups, Barentu and Borana, expanded in adjacent directions. The Barentu primarily moved into areas now known as Hararghe, Arssi, Wallo, and northern and eastern Shawa, while the Borana expanded into regions that include western Shawa, Kaffa, Gamu Gofa, Sidamo, Illubabor, Wallaga, and the Gibe areas. They achieved this expansion through their superior cavalry, surprise tactics, and the effective organization provided by the gada system (Jalata, 2019).

Music functions not only as entertainment but also as a powerful medium for storytelling, fostering social cohesion, and expressing cultural identity. It is often reserved for special occasions when singers aim to convey their innermost feelings (Bessa, 2013). With a diverse range of musical styles, instruments, and vocal practices, Oromo music embodies the community's values and experiences (Tafa, 2019). Oral traditions, including songs and storytelling, are vital components of Oromo culture and identity. This music offers unique insights into how the Oromo value and cultivate their musical arts within their multicultural context and beyond (Abebe, 2021).

Throughout history, the Oromo have encountered numerous challenges, such as marginalization, political repression, and cultural assimilation (Walga, 2021). Nevertheless, they have successfully preserved their distinctive cultural practices, including their unique musical traditions. Oromo music acts as a vehicle for oral history, storytelling, and social commentary (Abebe, 2021).

Considering the historical and cultural importance of the Oromo tribe's music, it is vital to study and document their musical styles comprehensively. Oromo music has become an essential means of preserving oral history, fighting against ignorance, and acting as a catalyst for the liberation of the Oromo people, who have faced the severe consequences of civil conflict and famine.

Activist musicians within the Oromo community have been instrumental in reclaiming and revitalizing their identity, which has long been suppressed under oppressive conditions (Proceedings of the 2008 Oromo Studies Association Annual Conference, 2009). By exploring the nuances of their musical traditions, we can gain valuable insights into the collective memory, cultural practices, and historical challenges faced by the Oromo people. The Afran Qalloo musical and cultural band, established in Dirree Dhawaa in the early 1960s, was the first organized Oromo group dedicated to preserving their culture and raising awareness about the various forms of oppression experienced by the community (Bessa, 2013).

In recent years, the rise of machine learning has created new opportunities for analyzing and classifying music within diverse cultural contexts(Holland & Fiebrink, 2019). However, there is still a significant gap in systematic studies focused on classifying the unique music styles of each Oromo tribe. This research aims to address this gap by utilizing machine learning techniques to categorize and analyze the distinct features of Oromo music.

The main objectives of this study are to classify the various styles of Oromo music, identify the unique characteristics of each style, and contribute to the preservation of this rich cultural heritage. By focusing on these objectives, this research aims to illuminate the complexities of Oromo music and its importance within the broader context of Oromo culture. In conclusion, examining the music styles of the Oromo tribe is essential for gaining insights into their history, cultural practices, and collective identity.

1.2 Statement of problems

Cultural history of the Oromo nation did not get the proper scholarly attention. It is commonly distorted or biased and from all cultural themes, music is much neglected (Degu, 2021).In addition to its unique dance forms, Oromia boasts a diverse array of musical styles in each tribe with their own rhythm. Even for those born and raised in Oromia, pinpointing the origin of a song's melody can be challenging due to the similarities among songs rhythm and the variety present within this Oromo traditional music, which reflects the multitude of cultural groups. For outsiders, distinguishing between these styles may be even more difficult.

Globally, numerous studies have explored music classification based on various criteria, including style (Tzanetakis & Cook, 2002), mood or emotion (Misron et al., 2014), human body gestures(F. Zhang et al., 2016), human-computer interaction (Panwar & Mehra, 2011),and suspicious activity detection (Tripathi et al., 2018). Many of these studies have primarily relied on video attributes for classification (Dhakal, 2019). (Lansdown & He, 2019)note that few studies have utilized auditory attributes for this purpose but no one focus on Oromo tribe music.

In Ethiopia, some researchers have treated Oromia as a monolithic entity, overlooking the significant diversity in dance and music styles across different Oromo zones. This oversimplification presents a flawed perspective. For instance, research conducted by (Mekonnen et al., 2016) on classifying Ethiopian traditional dance has categorized regions such as Tigray and Oromia as a single unit, rather than examining the distinct tribes within these areas.

Generally, the classification of Oromo music, a rich and diverse cultural heritage, presents significant challenges due to the unique audio rhythms and stylistic variations inherent in the styles. Understanding which Music features are most critical for effectively distinguishing between different styles of Oromo music is essential for developing accurate classification systems. Furthermore, the application of machine learning techniques in this context raises questions about the most suitable algorithms and methodologies for achieving reliable results.

Despite the growing interest in music classification using machine learning, there remains a gap in research specifically focused on Oromo music. This gap not only limits the understanding of the cultural nuances embedded in the music but also hinders the development of tools that could benefit music preservation, education, and appreciation within and beyond the Oromo community.

In contrast, our research aims to explore the music of each individual tribe in Oromia, focusing on auditory features to classify Oromo music based on the distinct Rhythmic of each tribe. To our knowledge, this study is the first to classify Oromo tribal music solely using auditory attributes, as no prior research has specifically addressed the classification of Oromo music.

Music plays a crucial role in people's lives, fostering connections among individuals with similar interests and acting as a unifying force within communities (Sacks, 2007). Communities can often be recognized by the types of songs they produce and enjoy, as different groups exhibit unique musical preferences. A key characteristic that distinguishes one piece of music from another is its style (Raj et al., 2021). Each Oromo tribe has its own distinctive musical rhythm and style. Despite the growing interest in employing machine learning techniques for cultural analysis in different Countries Music, there remains a significant gap in understanding which specific algorithms are most effective for classifying the diverse musical rhythm of each Oromo tribe.

Existing studies often neglect the unique characteristics of these music styles, treating the Oromia region as a single category for classification (Mekonnen et al., 2016). This oversight restricts the potential for accurate classification and fails to reflect the rich diversity within Oromo music. Therefore, it is essential to explore and identify the most effective machine learning algorithms for classifying the music styles of Oromo tribes, ensuring that the distinct attributes of each rhythm are recognized and analyzed. This research aims to address this gap by systematically evaluating various machine learning techniques to assess their effectiveness in classifying Oromo tribal music.

1.3 Research Question

By doing this research work, we will answer the following questions:-

1. Which audio features are very important for Oromo music classification
2. How can machine learning techniques be applied to classify Oromo music based on its style?
3. Which machine learning algorithms or techniques are most effective for classifying Oromo tribe music styles?

By addressing these issues, this research aims to contribute to the fields of musicology, and machine learning, ultimately fostering a deeper understanding and appreciation of Oromo music and its classification challenges.

1.4 Research Objective

This study's objectives are classified into two main categories: general objectives and specific objectives. The details for each category are presented below.

1.4.1 General Objectives

The primary aim of this study is to build a model that can effectively classify the music styles of the Oromo tribe, utilizing machine learning techniques.

1.4.2 Specific Objectives

1. Review literature on music genre classification.
2. To identify audio features suits for oromo music classification
3. To apply machine learning algorithms to classify Oromo tribe music based on its rhythm and style.
4. To compare and evaluate the performance of different machine learning algorithms in classifying Oromo music styles.

1.5 Motivation

Oromo music is incredibly diverse and carries substantial cultural significance, serving as a vital expression of identity for the Oromo people (Abebe, 2021). This complexity underscores the need for a systematic and organized approach to categorize the various styles and elements within Oromo music. Cultural history of the Oromo nation did not get the proper scholarly attention. It is commonly distorted or biased and from all cultural themes, music is much neglected (Degu, 2021) .Given its richness and variety, there is a clear demand for a structured system to effectively understand and organize this musical heritage. Music plays a crucial role in the lives of the Oromo people, fostering connections and serving as a medium for cultural expression (Abebe, 2021). However, the challenges of identifying the origins of melodies due to similarities among songs from different tribes highlight a significant gap in existing methodologies. By utilizing machine learning techniques, we can leverage advanced analytical tools to accurately classify these distinct musical styles.

Despite the increasing interest in applying machine learning to cultural studies, there remains a significant lack of focused research on the classification of Oromo music styles. Many existing studies often overlook the unique characteristics that define the music of individual tribes, using generic algorithms that fail to capture the intricacies of Oromo musical forms. This research seeks to fill this gap by exploring which machine learning algorithms are most effective for classifying the diverse styles of Oromo music.

By systematically assessing these techniques, we intend to develop a customized approach that acknowledges and honors the distinctive auditory features of each tribe's music, ultimately contributing to the preservation and celebration of Oromo cultural identity. Furthermore, the classification of Oromo music is essential for fostering cultural awareness and appreciation both within Ethiopia and globally. In Ethiopia, many researchers have unfortunately regarded Oromia as a homogeneous entity, overlooking the rich diversity of dance and music styles across its various zones.

This oversimplification does not adequately recognize the unique cultural expressions of individual Oromo tribes. For example, studies like those conducted by (Mekonnen et al., 2016) have categorized Oromia as a one entity without exploring the specific characteristics of its distinct tribes. In contrast, our research aims to address this gap by focusing on the music of each individual tribe in Oromia, thereby promoting a deeper understanding of their music cultural heritage.

1.6 Research scope

The scope of this research centers on the classification of music styles from five Oromo tribes. While the study primarily focuses on a specific selection of music from these tribes, it also acknowledges the broader diversity present within the Oromo musical tradition. The research targets the classification of five distinct Oromo music styles: Shewa, Jima, Wollega, Hararge, and Borana.

Each of these styles features unique musical characteristics and drum patterns (rhythms) that make them well-suited for classification analysis.

The study examines the effectiveness of various machine learning algorithms in classifying Oromo music rhythmic styles. This includes the implementation and evaluation of supervised learning techniques such as K-nearest Neighbor, Random Forest, Support Vector Machines, and other ensemble techniques. The selection of these methods enables a thorough analysis and comparison of their performance in classifying the music styles of Oromo tribes. The research takes into account several relevant factors, including dataset size and diversity, audio feature extraction, and potential biases within the dataset, such as imbalances in sample audio data. It is important to note that the scope of this research is centered on the application of machine learning techniques for classifying Oromo tribe music styles. While the study recognizes the broader cultural and sociological dimensions of Oromo music, the primary focus remains on the technical aspects of classification through machine learning algorithms.

1.7 Significance of Study

The research on classifying Oromo tribe music styles is significant for several reasons. By accurately identifying and categorizing these music styles, the study contributes to the preservation and documentation of Oromo music and its cultural importance. The findings can aid in the creation of comprehensive archives and databases that organize and catalog the diverse styles within Oromo music, ensuring their accessibility for future generations. This preservation effort is vital for maintaining and celebrating the cultural heritage of the Oromo tribe.

The findings of this research can play a crucial role in cultural preservation efforts, not only for Oromo music but also for the wider Oromo culture. By accurately classifying and organizing Oromo music styles, the study contributes to the preservation and promotion of cultural practices, rituals, and storytelling traditions linked to these styles. This can positively impact Oromo communities by fostering cultural pride, revitalizing traditional practices, and encouraging the transfer of knowledge across generations.

As a General this study brings important benefits such as:-

- Ensuring the transmission of musical information to future Oromo generations.
- It also holds significance for music experts and composers who wish to incorporate these technologies into their artistic works, providing them with valuable tools and benchmarks for their research.
- The study has pedagogical significance in supporting the teaching and learning process in music schools, enabling better understanding and application of Oromo music history.
- Lastly, it contributes to the development of Music Information Retrieval (MIR) systems, advancing the field and its potential applications.

1.8 Contribution

The first significant contribution of this research is the creation of a comprehensive dataset specifically designed for Oromo tribe music recordings. Assembling and curating a high-quality dataset is essential for the success of any machine learning project. In this study, careful efforts were made to collect a diverse array of Oromo tribe music recordings. These recordings were meticulously organized, annotated, and validated to ensure their accuracy and reliability. The resulting dataset serves as a valuable resource for training and evaluating machine learning models in the classification of Oromo tribe music styles, allowing researchers and practitioners to achieve more accurate and robust outcomes.

Another significant contribution of this research is the identification and extraction of informative audio features specifically tailored for classifying Oromo tribe music styles. To effectively differentiate between the various music styles within the Oromo community, it is essential to capture and analyze relevant characteristics from the audio signals. A comprehensive investigation was carried out to identify key audio features that provide discriminative information related to the different styles of Oromo tribe music.

These features were meticulously extracted using advanced signal processing techniques. The resulting audio features offer valuable insights and representations that enhance classification accuracy and contribute to a deeper understanding of the distinct musical styles within the Oromo tribe. Additionally, the development of a robust machine learning model specifically designed for classifying Oromo tribe music represents another major contribution of this research. By leveraging the prepared dataset and the extracted informative audio features, a sophisticated machine learning model was carefully constructed to effectively analyze and classify the music styles.

1.9 Limitations of study

In any research endeavor, acknowledging limitations is crucial as it provides a clearer understanding of the study's scope and the potential impact on the results. One significant limitation of this research is the restricted focus on only five tribes within the Oromia region due to the unavailability of audio data for other tribes in online video and audio streaming platforms. While these five tribes represent a portion of the diverse musical traditions in Oromia, the findings may not be fully generalizable to the broader spectrum of Oromo music. This limitation may affect the comprehensiveness of the classification model, as variations in musical styles, instrumentation, and cultural context across the additional tribes could introduce nuances that are not captured in the current study. Another limitation relates to the feature extraction process employed in the analysis. Additionally, the reliance on publicly available audio data may lead to inconsistencies in quality and representation and we have used ML rather than DL in case we didn't get more Data needed for Deep learning.

1.10 Organization of the Thesis

The thesis is structured as follows and consists of four chapters, excluding chapter one above. In Chapter 2, various approaches that can be used for Music classification tasks are described well, along with related publications. The connected works present several literary genres that are relevant to our work. Chapter 3 explains the approach taken and provides details on the data used, preprocessing steps and feature extraction to train and evaluate the classifiers. An overview of the employed classification algorithms and the specifics of their implementation make up the majority of this chapter. We discussed the work's analysis and experimental findings in Chapter 4. The outcome of the implemented algorithms' testing and training is shown. Analysis is done based on the results gathered to determine which method works best for Oromo audio Music classification purpose. Lastly, Chapter 5 presents the conclusion and recommendation for upcoming research projects.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

In this chapter, we present an examination of existing literature and an analysis of related work, specifically focusing on the classification of music. This review includes fundamental music theories, the unique qualities and characteristics of Oromo music, and various aspects of audio analysis. Our comprehensive overview provides clear insights and guidance for identifying the key requirements involved in performing music classification using machine learning techniques, including feature selection and classification algorithms. Additionally, a brief explanation of machine learning algorithms is included to enhance understanding of their application in this context.

2.2 Overview of Music and its classification

Music plays a significant role in daily life. Globally, it manifests in a variety of forms and genres. Individual musical tastes differ widely (Retta et al., 2023). Music Information Retrieval (MIR) is a highly interdisciplinary field of research that focuses on extracting information from music and its various applications. Typically, music refers to audio content, but its definition also encompasses other types of musical information, such as lyrics, metadata, or user listening habits. The classification of music by genre is the most effective method. This classification is primarily performed through manual labeling. Such categorization is essential for music recommendations, as individuals generally prefer listening to music from their favored genres. Numerous music applications, including Spotify and Sound Cloud, utilize genre classification to recommend tracks to their users.

Classifying music genres has emerged as a significant area of research due to its impact on the development of music information retrieval systems and the enhanced management of musical records. This classification enables a comprehensive analysis and better understanding of music content. Describing a musical genre manually requires considerable time and effort from experts (Terefe, 2019). The effectiveness of a given set of features varies based on the specific application. Consequently, a primary challenge in creating audio classification models lies in developing descriptive capabilities tailored to particular uses. In essence, audio conveys much about a clip's

mood, musical elements, noise levels, and tempo, allowing the human brain to classify sounds based solely on audio cues (Muller et al., 2011). Music genres can be identified by examining numerous characteristics within the music. Thus, automating the classification of musical genres poses a complex challenge. Nonetheless, distinct patterns can be observed within the same genre by exploring the relationships among various features.

2.3. Audio representation

An audio signal refers to a signal that carries information within the range of audible frequencies. Audio representation involves extracting properties or features of the audio signal that reflect its composition in both the temporal and spectral domains, as well as its behavior over time (Choi et al., 2017).

The research by (Pielemeier et al., 1996) highlights that audio time-frequency representation provides more detailed insights into audio signals. There are millions of music data entries in digital databases, each representing music in various forms. These can include text formats like song lyrics, visual formats such as album posters, and audio formats used in recordings. Another representation is the Music Instrument Digital Interface (MIDI), which encompasses different ways to represent music and audio that are essential for studies and research in music information retrieval.

2.4. Audio signal processing

Audio signal processing centers on computational methods aimed at intentional manipulation of acoustic signals or sounds which involves enabling machines to interpret sounds from various events (Dhanasingh & Hochmair, 2021). Sounds originate from a variety of sources, including humans, musical instruments, animals, and other objects (Turchet et al., 2023). Audio signal processing plays a crucial role in several application areas, such as music, speech recognition, and environmental sound detection for purposes like surveillance, information retrieval, and communication. Similar to humans, machines can recognize audio signals, which facilitates decision-making based on predefined rules and criteria. To understand and analyze these sonic properties, specific audio signal processing techniques are essential.

2.4.1. Fourier Transform

The Fourier Transform is a mathematical technique that transforms a time-domain signal into its frequency-domain representation. This is crucial for analyzing the frequency content of audio signals. The Fast Fourier Transform (FFT) is an efficient algorithm that computes the Fourier Transform, allowing for quick analysis of audio data. By converting audio signals into their frequency components, practitioners can identify dominant frequencies, harmonics, and noise, facilitating tasks like spectral analysis, sound synthesis, and audio effects design.

2.4.2. Time-frequency analysis techniques,

Time-frequency analysis techniques, such as the Short-Time Fourier Transform (STFT) and Wavelet Transform, are essential for examining signals whose frequency content varies over time. STFT divides a signal into short segments and applies the Fourier Transform to each segment, providing a time-varying frequency representation. This is particularly useful for analyzing transient sounds, such as percussion. Wavelet Transform offers a more flexible approach by using wavelets of varying lengths, enabling multi-resolution analysis. This adaptability makes it effective for capturing both high-frequency details and low-frequency trends in non-stationary signals.

2.4.3. Machine learning techniques

Machine learning techniques are increasingly applied in audio signal processing to automate and enhance various tasks. These include audio classification, where algorithms are trained to identify different sounds or genres, and source separation, which isolates individual components from a mixed signal, such as separating vocals from instruments. Machine learning can also improve noise reduction and audio enhancement by learning from large datasets to better understand audio characteristics. As technology advances, these techniques are becoming vital for developing intelligent audio applications and innovative sound processing tools.

2.5. Music Classification

Music classification involves categorizing music into distinct styles or types based on specific characteristics. This practice has evolved over time, initially relying on manual classification methods employed by musicologists and ethnomusicologists. Traditional classification often considers elements such as instrumentation, rhythm, and cultural context, which can be subjective and open to interpretation (Tzanetakis & Cook, 2002).

In recent years, technological advancements have transformed music classification, enabling more objective and efficient methods. Machine learning algorithms can analyze large datasets of audio features, facilitating automated classification based on learned patterns. This shift not only improves accuracy but also allows researchers to explore music classification on a larger scale, accommodating the complexities of diverse musical styles, including those of the Oromo (Schedl et al., 2014).

2.5.1. Traditional Music Classification

Music has persisted from its inception to the present day, with traditional music exhibiting certain fundamental characteristics. For instance, it is often transmitted through oral traditions, passed down from generation to generation. Additionally, traditional music reflects the cultural identity of a nation and is closely linked to historical and personal events. Various songs and musical forms have been performed as part of customs and traditions over extended periods, spanning multiple generations. This type of music holds significant cultural value and is regarded as a national heritage, providing insights into a country's history, traditions, and collective beliefs. It serves as a means for individuals to gain a deeper understanding of a community and its cultural fabric. In essence, music classification is the process of assigning different musical works to distinct categories based on a variety of criteria. This is done by Experts or musicologists listen to and analyze pieces of music to categorize them based on established criteria.

While musical characteristics such as artist names, song titles, album titles, and vocal styles are commonly used, most contemporary music classification studies also consider additional musical features through digital signal processing. The term used to categorize specific musical styles that adhere to a shared custom or set of guidelines is "style" or "genre" (Song et al., 2017).

Because music is an artistic medium, these classifications are often arbitrary and contentious, with certain styles potentially merging. In our case, for example, the music from the eastern and western Wollega Zone is nearly identical and belongs to the broader Wollega traditions. Therefore, the music from the eastern, western, and Kellem Wollega zones is grouped together under the Wollega tradition, as they share similar musical styles and dance developments.

The classification of music across different outlets, from physical stores to streaming services, relies heavily on style and genre designations. This categorization significantly influences the development of music information retrieval systems and enhances methods for managing musical records. As a result, the classification of music by style, mood, or emotion is rapidly emerging as a leading research topic (Bahuleyan, 2018). This approach enables a more comprehensive analysis and clearer description of the musical elements involved.

2.5.2 Automatic Music Classification

Automated Music Classification is a computer process that assigns musical records to interest-based groups based on their representative musical characteristics (Chai, 2005)s. Since most categorization relies on quantized or digital records, this automated classification is essential for achieving accurate results. To effectively categorize musical files, it is necessary to characterize these entities in terms of their distinctive musical properties, which are numerically recorded. Criteria for categorizing musical qualities can include timbre, texture, rhythmic content, pitch content, or combinations of these elements. All operations involved will utilize computer technology. Digital music classifications can take various forms, such as artist filtering, style classification, and mood classification.

To create a robust classification system for these tasks, several fundamental procedures must be followed: gathering sample audio, performing preprocessing to extract relevant features, selecting appropriate tools and techniques for classification, and so on. The system will be developed using the extracted features, with classification executed through the chosen tools and techniques.

2.6 Methods for Classifying Music

2.6.1. Rule based Feature extraction

Rule-based feature extraction is a method used to derive relevant features or characteristics from data based on predefined rules or criteria. In the context of music classification, this approach involves creating specific rules or guidelines to identify and extract meaningful features from music signals or related data. These rules can be informed by domain knowledge, expert insights, or empirical observations, defining the conditions or patterns that must be met for a particular feature to be extracted. Rule-based feature extraction may analyze various aspects of music, including audio properties, rhythm, melody, tonality, and structural elements. In contrast, perceptual features such as loudness, brightness, pitch, timbre, and rhythm are subjective characteristics closely linked to the perceived qualities of sound (T. Zhang & Kuo, 2013).

2.6.2 Machine learning techniques

Through data analytics, machine learning enables computers to learn from experience in a manner akin to humans and animals. Instead of relying on a model based on predefined equations, it utilizes computational methods to extract insights directly from data (Nithya & Saravanan, 2018). Using this learning approach, computers undergo various training processes to identify naturally occurring patterns in sample data, enhancing their understanding. Machine learning algorithms typically consist of two fundamental phases: classification and regression, and they can be categorized into two types: supervised and unsupervised. Supervised algorithms work with labeled data, while unsupervised algorithms deal with unlabeled data, training models based on existing input and output data to predict future outcomes(Zhou et al., 2017). For

instance, unsupervised algorithms can classify unsorted data by identifying similarities and differences. Additionally, many machine learning systems, such as active learning and transfer learning, incorporate semi-supervised methods to enhance their effectiveness.

For various tasks, including recognition and classification, researchers in music style classification utilize a range of machine learning algorithms (Bahuleyan, 2018). Depending on the specific task, they select the most suitable machine learning methods, such as Naive Bayes (NB), K-nearest Neighbor (KNN), Support Vector Machine (SVM), and others, for both manually created and automatically extracted features.

2.7. Machine learning Classifiers

2.7.1 Support vector machine (SVM)

Support Vector Machines (SVMs) are a type of supervised learning algorithm used in machine learning. In classification problems, it is commonly observed that data points from the same class tend to cluster together on graphs representing their feature values. To make predictions for new data points, an n−1 dimensional hyperplane can be employed to separate the data, where n represents the number of features. If a new data point falls on one side of the hyperplane, it is predicted to belong to one class; if it falls on the other side, it is predicted to belong to a different class.In our case, Support Vector Machines (SVMs) will classify music by following a series of steps. First, relevant features such as average beat frames, Mel-frequency cepstral coefficients (MFCC), and spectral features are extracted from the music data. The selection of features depends on the specific classification task and the available data. Once the features are extracted, a labeled dataset is created, with each music sample linked to its corresponding music style label. The dataset is then divided into a training set and a test set. The SVM algorithm is trained on the training set, where it learns to identify an optimal hyperplane that separates different music styles based on the extracted features.

The objective is to maximize the margin between the classes, thereby enhancing the classifier's ability to generalize to unseen data. Once trained, the SVM can predict the music style of new, unseen samples by assigning them to the appropriate side of the hyperplane. The accuracy of the SVM model is evaluated by comparing its predictions on the test set to the true style labels. Through iterative refinement of the features and the model, the classification performance can be improved.

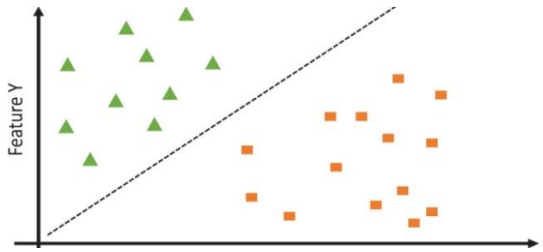


Figure 1: how Support Vector Machine work

2.7.2. Random Forest

Random Forest is an ensemble learning method that aggregates predictions from multiple decision trees. It is based on two key principles: (1) each decision tree is trained using a subset of the training samples through a process known as bootstrap aggregation (or bagging) (Breiman, 1996), and (2) each tree makes predictions using a random subset of the features (Amit & Geman, 1997). The final predicted class for the Random Forest is determined by a majority vote from the individual classifiers. In our case, Random Forest is chosen for its robustness and ability to handle diverse data, making it effective for capturing complex patterns in audio features. It is a widely used machine learning algorithm suitable for music classification. The process of applying Random Forest for music classification involves several steps. First, relevant features such as average beat frames, Mel-frequency cepstral coefficients (MFCC), and spectral features are extracted from the music data. These features capture important characteristics of the music that help differentiate between various styles. Once the features are extracted, a labeled dataset is prepared, linking each music sample to its

corresponding style label. The dataset is then divided into a training set and a test set.

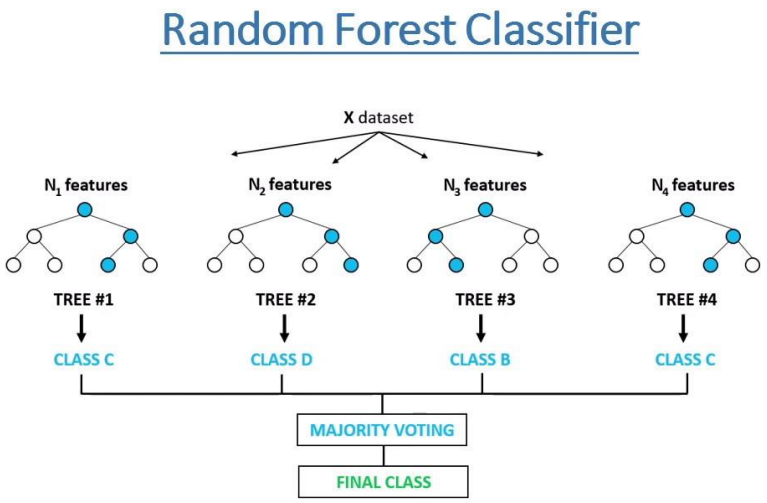


Figure 2 How Random Forest Classifier works (Khushaktov, 2023)

Next, the Random Forest algorithm is trained on the training set. It creates an ensemble of decision trees, with each tree trained on a random subset of the features and data samples. During training, each decision tree learns to make predictions based on different combinations of features. This diversity enables the Random Forest model to capture complex relationships and patterns within the music data. Once the Random Forest model is trained, it can classify the style of new, unseen music samples. For a given input sample, each decision tree in the ensemble independently predicts the style, and the final prediction is determined through majority voting. The style that receives the most votes from the individual decision trees is selected as the predicted style for the input sample. To evaluate the performance of the Random Forest model, its predictions on the test set are compared to the true style labels. Various evaluation metrics, such as accuracy, precision, recall, and F1 score, can be calculated to assess the model's effectiveness. By iterating and refining the feature selection, training, and evaluation processes, the Random Forest model can be enhanced to more accurately classify music into different styles.

2.7.3. Support Vector Machine

The Support Vector Machine (SVM) is a supervised classification technique that identifies the hyperplane with the largest margin between two classes of data and reason for the popularity of SVM is its good generalization performance to classify unseen data (Kinnunen & Li, 2010). When the data cannot be separated linearly in the feature space which is often the case it can be transformed into a higher-dimensional space using a Mercer kernel, K(·) (Mandel & Ellis, 2005).

2.7.4 K-Nearest Neighbors

The k-Nearest Neighbors (KNN) algorithm is a simple yet effective method for classification (Guo et al., 2003). It provides a straightforward approach by leveraging feature similarity to classify data points. In contrast, the ensemble method that combines Random Forest (RF) and KNN aims to capitalize on the strengths of both algorithms, resulting in improved accuracy for classification tasks.

2.7.5. XGBoost Boosting

Boosting is an ensemble classification technique that combines multiple weak learners, such as decision trees. Unlike Random Forests, boosting algorithms are trained sequentially through a process known as forward stage wise additive (Hastie et al., 2009). According to (Bahuleyan, 2018), XGB stands for eXtreme Gradient Boosting, which is a specific implementation of boosting designed for efficient and parallelized model training.

2.9.6 Gradient Boosting

Gradient Boosting is an ensemble machine learning algorithm known for its ability to build strong predictive models through iterative learning, focusing on correcting the errors of previous models. Together, these algorithms offer a comprehensive framework for analyzing and classifying Oromo music, leveraging diverse methodologies to achieve optimal results.

In our work, relevant features such as average beat frames, Mel-frequency cepstral coefficients (MFCC), spectral features, and tempo are extracted from the music data. These features capture essential musical aspects that help distinguish between different styles. Once the features are extracted, a labeled dataset is created, associating each music sample with its corresponding style label. During the training process, the decision tree learns a series of rules or conditions that guide the classification of new, unseen music samples.

Recent developments in music processing have primarily focused on handling audio signals. This field has expanded the capacity to access, analyze, manipulate, and create music. Given the complexity and diversity of musical forms, research has explored various aspects of music, including style, instrumentation, dynamics, tempo, rhythm, and timbre. Music signals are inherently multimodal, often combining audio with video and images, and incorporating a wide variety of sound sources.

Due to the vast amount of data involved, music processing presents significant challenges and is closely related to fields such as machine learning, multimedia signal processing, and audio and acoustic signal processing (Gan et al., 2008). Music transmissions differ from spoken language and other audio signals in their acoustic and structural properties.

Therefore, the musical pieces examined in this context are typically songs or compositions. Specifically, we first categorized different aspects of music audio according to multiple criteria, including abstraction level (from low to high), temporal scope, and frequency versus time domain.

2.8. Audio Feature Extraction

Computational features that aimed to capture some of the facets of human music perception were distinguished using a variety of dimensions. These characteristics were categorized by:

2.8.1 Temporal Scope

In music, three levels of characteristics are distinguished: immediate, segment-level, and global aspects. Segment-level features represent a specific duration of consecutive sounds and can be calculated using one of two methods: a semantically relevant definition of a segment (such as a musical phrase or the chorus of a song) or a predefined time window (such as 5 or 10 seconds). In contrast, instantaneous characteristics are computed for a particular moment in time and typically span only a few tens of milliseconds. This layered approach allows for a comprehensive analysis of musical elements across different time scales.

2.8 2 Domain-Based Features

Audio signals can be represented in two ways: the time domain and the frequency domain. The time domain representation shows the signal's amplitude at each point in time, reflecting the value of the signal at each sampled instant. In contrast, the frequency domain representation is achieved through a Fourier transform, which decomposes the signal into multiple waves oscillating at different frequencies. This transformation provides insights into the various frequency components that make up the audio signal.

2.8.3 Spectrogram

A spectrogram visualizes the power (or "loudness") of a signal over time across different frequencies that make up a waveform. It is essentially the Short Time Fourier Transform (STFT) of an audio signal. Most spectrograms depict the frequency and temporal components of an audio signal (Nirmal & Mohan, 2020).Spectrograms are typically represented as two- or three-dimensional graphs, with a fourth variable indicated by color (Hajarolasvadi & Demirel, 2019).

2.9 The Oromo

The Oromo, Ethiopia's largest ethnic group, boasts a rich tapestry of cultural practices. A key event that highlights Oromo traditions and beliefs is the Irreecha festival, a grand celebration that attracts millions of participants. During this festival, Oromo individuals proudly showcase their cultural heritage (Debele, 2017). Traditional clothing includes garments like the Uffata, a toga-like robe typically worn by men, often in white. Women wear a skirt known as a wandabo and embellish their attire with textiles such as qollo and sadetta, which are cotton fabrics traditionally hand-spun and woven. They also don robes made from leather or animal skins (Einashe, 2016). The Oromo people actively engage in customs that distinguish them from other Ethiopian ethnic groups, thereby maintaining a unique cultural identity. They celebrate their identity and preserve their rich cultural history, despite variations in geography and religious beliefs. The Oromo community places significant value on music, which serves as a powerful means of expressing joy, honoring heroes, and conveying grief and mourning. Music is particularly vibrant in public gatherings, on the battlefield, during agricultural activities, and throughout harvest seasons. It plays a vital role in promoting cultural expression, preserving identity, and negotiating values within the community. Each tribe in Oromia is characterized by its unique drum beats and rhythms. In our study, due to limitations in data availability, we focus on five specific Oromo tribal music styles: Shewa, Wollega, Hararge, Jima, and Borena.

2.9.1 Shewa Oromo

The Shewa Oromo, a subgroup of the Oromo people living in central Ethiopia, have a distinctive dance style that reflects their cultural identity. The Shewa Oromo dance is marked by unique costumes and choreography. Women wear striking two-piece outfits made from leather and adorned with shells, while men perform with sticks and don lion's manes on their heads. A particularly intriguing aspect of the Shewa Oromo dance rhythm is the remarkable agility and speed of the women's neck movements. They execute rapid and precise rotations, creating a captivating visual spectacle. The women transition from a slow, circular motion of the head to sudden bursts of speed, achieving an impressive pace (Ethiopia, n.d.).

2.9.2 Hararghe Oromo

Historically, Hararghe was recognized as a political entity known as Awraja under previous Ethiopian regimes. In the current political structure, Hararghe is divided into Western and Eastern Hararghe, which, along with seventeen other regions, form the 19 zones of the Oromia regional state. The Hararghe highlands cover approximately 1,500,000 hectares, stretching from Galamso and Chiro Mountain in the west to Kundudo Mountain in the east. Located about 400 kilometers east of Addis Ababa, the capital city of Ethiopia, Hararghe shares its borders with Djibouti and Hargeisa to the west. Traditionally, the elders of this region perform a song called Mirriyisaa (Getahun, 2019). Additionally, one of the most renowned dances rhythm among the Hararghe people is called Shagoye, which is performed by both men and women. During this dance, it is customary for individuals of different genders not to form separate couples, but instead to engage in a dynamic interaction across opposing rows.

2.9.3 The Jimma Oromo

As previously mentioned, the Jimma are one of the four sub-regional Oromo groups that comprise the southwestern Oromo population in Ethiopia. The name Jimma refers to both the people and the land they inhabit, located west of the Gibe River and east of the Gojeb River. This region is now recognized as the Jimma administrative zone of Oromia. According to a well-known historical account, the term "Jimma" originated from a man named Jimma Sirba, who was the first to settle in the area. While the name now encompasses all Oromo people in the Gibe region, it originally referred specifically to five clans, known as "gosa" in the local dialect. These clans Qore, Harsu, Lalo, Bilo, and Badi were descendants of Jimma Sirba and were the first to establish themselves in the Jimma area during the 19th century under the leadership of Jimma Abba Jifar (Alemu, 2012). This tribe has their own music rhythm which is called " jigjigse and Gadee" based on physical movement of dancer.

2.9.4 Borena Oromo

The Borana Oromo are part of the larger Oromo national group, which constitutes about 40% to 50% of Ethiopia's population. They inhabit a peripheral region that includes the strategic borders of Ethiopia, Kenya, and Somalia. Historically, this area has been significant since ancient times. However, following the conquest of the region by the Ethiopian imperial state in the last decade of the 19th century, the pastoral populations

of the Borana Oromo underwent significant political, economic, cultural, and ecological changes. One notable change was the boundary delimitation in 1907, which divided the Borana Oromo between Ethiopian and British-Kenyan jurisdictions (Duressa, 2022). This tribe follows Reggae Rhythm as their cultural rhythmic identification.

2.10. Related Work

In this part, we reviewed works pertaining to the classification of music works that addressed the issue of music classification using a different machine learning approach.

2.10.1 Machine Learning-Based Classification of Music

Machine learning-based classification of music utilizes algorithms to analyze audio features and categorize music into distinct styles. By employing techniques such as feature extraction, neural networks, and supervised learning, these systems can effectively identify patterns and similarities within musical compositions. This approach enhances the ability to classify music accurately and efficiently across various styles.

Researchers (Lansdown & He, 2019) compared various machine learning algorithms for automatically classifying song excerpts into their respective musical styles. They reviewed existing techniques and approaches related to feature engineering and available algorithms. To conduct their analysis, they created two distinct datasets from an online music archive, balancing the classes through under sampling and oversampling techniques. Each sample was analyzed using audio analysis libraries to extract fifty-four features.

The study involved two sets of analyses, comparing the average feature values and F-values for each style. Four classifiers were trained and tested on each dataset, including a neural network, support vector machine, random forest, and gradient boosting machine. The results indicated that the support vector machine algorithm was the most effective for this task, with Hip-Hop music being the style most accurately classified on average. Additionally, the study evaluated the entire process, encompassing dataset selection, feature selection, and project management aspects.

The automatic classification of audio signals into a hierarchy of musical styles was proposed by (Tzanetakis & Cook, 2002). At the top of this hierarchy, audio signals are categorized as either speech or music. Speech is further divided into categories such as male, female, and sports, while music is classified into ten primary styles: pop, metal, jazz, country, disco, hip-hop, classical, blues, rock, and reggae. Moreover, the classical category includes four subcategories (Choir, Orchestra, Piano, String Quartet), and the jazz category features six subcategories (Big Band, Cool, Fusion, Piano, Quartet, Swing). The authors note that while a specific style of music may lack precise definitions and boundaries, its members often share common characteristics in terms of instrumentation, rhythmic structure, and pitch content.

The authors proposed thirty elements to describe musical characteristics: five features for pitch content, six for rhythmic content, and nineteen for timbral texture. Of the nineteen features suggested for timbral texture, nine were calculated using the Short Time Fourier Transform (STFT), while the remaining features were derived from the Mel-Frequency Cepstral Coefficient (MFCC). To perform classification, they employed various statistical pattern recognition (SPR) classifiers to estimate the probability density function (PDF) for the feature vectors of each class. This work not only laid the groundwork for further research in music classification but also served as a pioneering effort in the field.

Building on this foundation, (Ren et al., 2010) sought to incorporate sequential temporal information into music style categorization, an aspect that earlier methods overlooked. The authors argue that music can be conceptualized as a series of sound events, highlighting the importance of temporal dynamics in classification. As a result, they proposed the song tokenization technique, which converts music into a series of discrete units. In this approach, they utilized acoustic segment models (ASMs) to transform the music into a sequence of tokens. The Hidden Markov Model (HMM) was employed to determine the variable length of these tokens. Subsequently, they integrated the temporal information of the music signal into the HMM.

By applying the music tokenization method, they effectively redefined the music style classification problem as a text classification challenge, which can be addressed using standard text information retrieval strategies. The dataset used for this study was Magnatunes.

Five styles used to categorize the music: rock, jazz/blues, electronic, metal/punk, and classical. The authors' experiment revealed that the sequential patterns (SPs) mined from the "Classical" class exhibited the strongest discriminative power, while the "Metal/Punk" and "Rock" classes were found to be quite similar, indicating a level of confusion between them. The results demonstrated that their proposed strategy, which incorporated sequential patterns, outperformed earlier methods that relied solely on local temporal features, achieving an accuracy of 74.19%.

In a related study, (Bahuleyan, 2018) focused on music genre classification using machine learning approaches, providing a method for automatically tagging songs in a user's library. This research explored the use of artificial neural networks alongside more traditional machine learning algorithms to achieve its objectives.

The initial approach employed a fully trained convolutional neural network, utilizing features derived from signals such as spectrograms, which can be treated as images. The methodological approach employs various machine learning techniques, including Random Forest and Logistic Regression, utilizing features extracted from the temporal and frequency domains of the audio signal.

Additionally, some researchers have explored the applications of different machine learning algorithms, such as k-Nearest Neighbor (KNN), Support Vector Machine (SVM), and Artificial Neural Network (ANN), in the realm of music style and mood classification detailed an audio classification approach using an artificial neural network, emphasizing the importance of audio feature extraction methods, such as Chroma-based features and Mel-frequency cepstral coefficients (MFCCs), to better understand the behavior of audio signals. The following table summarizes related work in this area.

As Ethiopia, Recent research on video classification has highlighted the richness of Ethiopia's diverse cultural heritage, particularly in relation to traditional music and dance (Moges, 2023). A study aimed at developing a classification model for Ethiopian traditional music videos addresses the challenges users face in accessing these cultural resources online. By employing Convolutional Neural Networks along with feature extraction methods such as Histogram of Oriented Gradients (HOG) and Optical Flow, the study successfully classified 80 video clips representing various ethnic groups, achieving an accuracy of up to 86% when combining features.

This work underscores the relevance of advanced machine learning techniques in organizing and preserving cultural expressions. While this study focuses on video content and focus solely on one region as one class, it parallels the need for effective audio classification systems, such as in the case of Oromo music that we are doing on, where similar methodologies for feature extraction and classification can be applied. By drawing from insights gained in video classification, particularly regarding feature extraction and dataset development, our research aims to enhance the understanding and categorization of Oromo musical styles, ultimately contributing to the preservation and appreciation of this rich cultural heritage of each Oromo tribe.

In recent years, there has been significant research focused on the automatic classification of music genres through sound analysis, particularly for applications in music information retrieval (MIR), data analysis, and transcription. As the volume of music data continues to grow, the challenges associated with indexing and retrieving this information have become more pronounced. Previous studies have primarily concentrated on the classification, identification, prediction, and differentiation of music genres within modern music services. One specific area of interest is the classification of Aquaquam zema, a traditional form of music associated with the Ethiopian Orthodox Tewahdo Church (EOTC) done by (Dagnew, 2023). This genre involves priests performing with a measured sound, emphasizing its cultural significance and its connection to rhythm and music as a secular art form. To address this, researchers have developed models capable of categorizing Aquaquam zema into distinct genres, such as Zimame, Qum (Tsinatsel), Meregd, Tsifat, and Amelales.

Utilizing audio data collected from Aquaquam bet, researchers have implemented preprocessing techniques, including audio segmentation and spectrogram generation, followed by the application of deep learning models, specifically Convolutional Neural Networks (CNNs). Results have demonstrated impressive accuracy rates, with training accuracy reaching 97.5% and test accuracy at 91.76%. This work underscores the potential of machine learning techniques in preserving and classifying traditional music forms, highlighting a path that could be mirrored in the classification of Oromo music genres.

Table 1 Summary of related work

AUTHOR	FEATURES	ALGORITHMS	DATASET	APPROACH	ACCURACY	YR
Ren et al.	Sequential temporal	KNN	Magnatunes	ML	74.19%	2010
K. Kumar	MFCC, Chroma	NN	FMA	ML	89.2%	2020
Lansdown & He	54features manually extracted	SVM	Created by researcher	ML	68%	2019
Tzanetikas &cook	STFT, MFCC	SPR	GTZAN	ML	61%	2002
Bahuleyan, 2018).	MFCC, chroma, spectral	LG, RF.XGB,SVM	Created by researcher	ML	89%	2018
(Moges, 2023)	145 feature Vector	CNN, HOG,OF	Created by researcher	DL	86%	2023
(Dagnew, 2023)	Spectrogram	CNN	Created by Researcher	DL	97.5%	2023

2.11. Research Gap

The existing literature highlights a notable gap in the automated classification of Oromo tribal music, primarily due to the lack of standardized methods and comprehensive datasets. While there has been some research on music classification in general, There is insufficient specialized research on Oromo music classification with appropriate auditor features and many existing studies rely on generic algorithms that fail to recognize the unique musical traits of individual Oromo tribes (Mekonnen et al., 2016). Like this, specific studies focusing on the unique characteristics of Oromo tribe music style classification using machine learning or recent subset of Ai technology are scarce.

This lack of attention results in a reliance on subjective categorizations that are inconsistent and potentially biased. Furthermore, the technological advancements in machine learning have not yet been effectively applied to this area, which presents an opportunity for innovative research that can contribute to both the fields of musicology and machine learning and also from the reviewed related work on music classification above, maximum accuracy gained from classifiers are 89% which may be taken as low. Addressing these gaps is essential for enhancing the understanding and preservation of Oromo musical heritage.

CHAPTER THREE: MATERIALS AND METHODS

3.1. Introduction

This chapter presents the methodology and materials utilized in the research study on classifying the music of the Oromo tribe using machine learning (ML) techniques. It outlines the research design, data collection and preprocessing, feature extraction, model selection, and evaluation metrics. In this section, we detail the methodology and steps we undertook to classify Oromo tribe music styles. The architecture of the proposed classification system is illustrated in the following diagram.

The methodology provides a systematic approach for achieving the research objectives and addressing the research questions. It combines various machine learning algorithms and techniques to classify five distinct styles of Oromo tribe music. The primary focus is on supervised learning algorithms and ensemble ML techniques for audio classification tasks. The overall methodology is outlined as follows:

3.2. Research Design

The research design describes the overall plan and structure of the study. In this research, an exploratory approach is adopted to analyze the relationship between Oromo tribe and music style using ML techniques. The study involves several stages, including data collection, preprocessing, feature extraction, model training, and evaluation.

3.3 System Architecture

The system architecture for the Oromo tribe music classification project is designed to facilitate the effective processing and analysis of audio data, leveraging multiple machine learning algorithms. At its core, the architecture consists of several key components: data acquisition, preprocessing, feature extraction, model training, and classification. Initially, audio files are collected .The preprocessing module then segments the audio into manageable clips after that relevant features, including Mel-Frequency Cepstral Coefficients (MFCCs), Root Mean Square (RMS), spectral features, tempo, and average beat frame extracted. These features serve as the input for various machine learning models, including Random Forest (RF), Support Vector

Machines (SVM), K-Nearest Neighbors (KNN), Ensemble methods combining RF and KNN, as well as advanced techniques like XGBoost and Gradient Boosting. The model training phase employs cross-validation to optimize hyper parameters and evaluate performance metrics such as accuracy and F1 score.

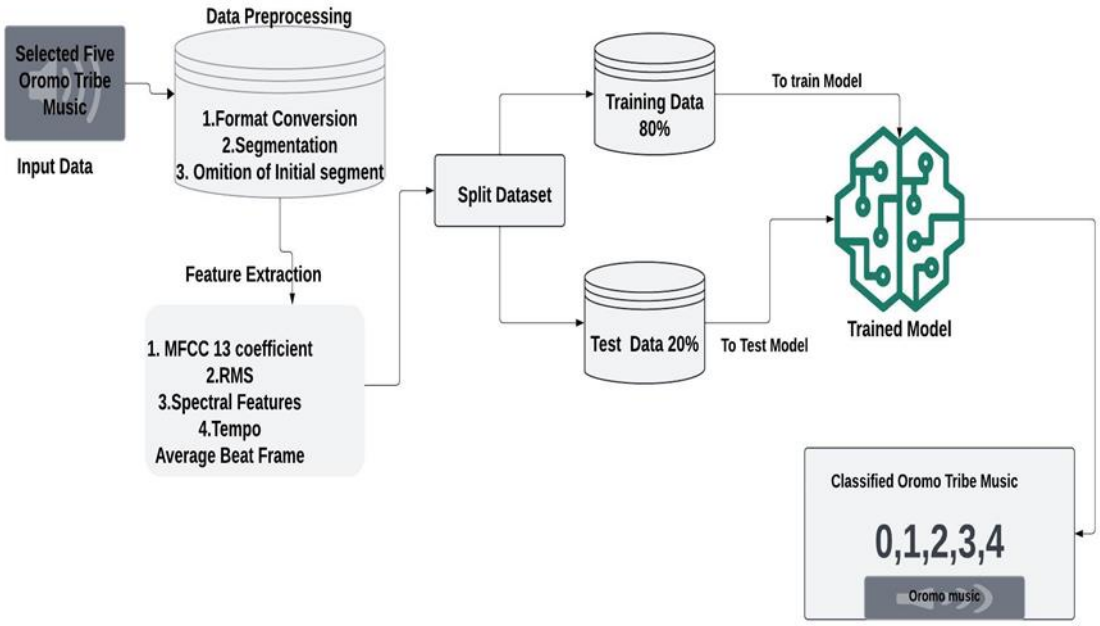


Figure 3 System architecture

3.4. Aim of Research

The primary objective of this research is to classify the diverse musical styles of the Oromo tribe using machine learning techniques based on numerical audio features. By extracting features from audio files converted from video sources, the study aims to identify distinct patterns and characteristics inherent to Oromo music. The research seeks to answer specific questions regarding the classification accuracy of different machine learning algorithms and aims to explore how cultural elements influence musical structure. Ultimately, this study aspires to contribute to the preservation and appreciation of Oromo music by providing a systematic approach to its classification.

3.5. Data Source

Data plays a crucial role in any research endeavor, particularly in the field of music classification, where the accuracy and effectiveness of algorithms depend heavily on the quality and diversity of the dataset. In our case, there was no publicly available dataset of audio data for each Oromo tribe music style. Since there was no prepared data for this task, we created our own dataset by downloading from a video-sharing platform. We used YouTube to gather data for the classification of Oromo tribe music styles. The initial step involved downloading video content, which we then converted to audio data. This audio was initially in MP3 format, requiring conversion to .wav format using audio and video processing tools called Cap Cut. This dataset was checked and labeled by Dambi Dollo university Music department Head and Folklore Department Head to relevant tribe. Our goal was to compile a dataset that accurately represents the diverse music styles of five the Oromo tribes.

For this study, we focused on five distinct Oromo tribe music classes, each contributing to our dataset. The first tribe Borena included all downloaded original audio files totaling 60 entries. After segmenting these files into 30-second clips, we obtained 590 segments, resulting in a total data size of 743 MB and an audio length of 4 hours and 55 minutes.

The second tribe, Hararge, consisted of 58 original audio files, which, after segmentation, yielded around 537 segments. This tribe's data totaled 676 MB with a cumulative audio length of 4 hours and 28 minutes. For the Jimma tribe, we downloaded 42 original audio files, resulting in 445 segments after processing. The total data size for Jimma was 560 MB with a time length of 3 hours and 43 minutes.

The Showa tribe contributed 60 original audio files, which, after segmentation, amounted to 682 segments. The total size for the Showa tribe's audio data was 859 MB, with an audio length of 5 hours and 42 minutes. Lastly, the Wollega tribe had 61 original audio files, resulting in around 562 segments after segmentation. This data totaled 708 MB with a cumulative audio length of 4 hours and 4 minutes. We have encode label of our dataset as the following table.

Table 2 Class name and their code

No	Class Name	Encoded as
1	Borena	0
2	Hararge	1
3	Jimma	2
4	Showa	3
5	Wollega	4

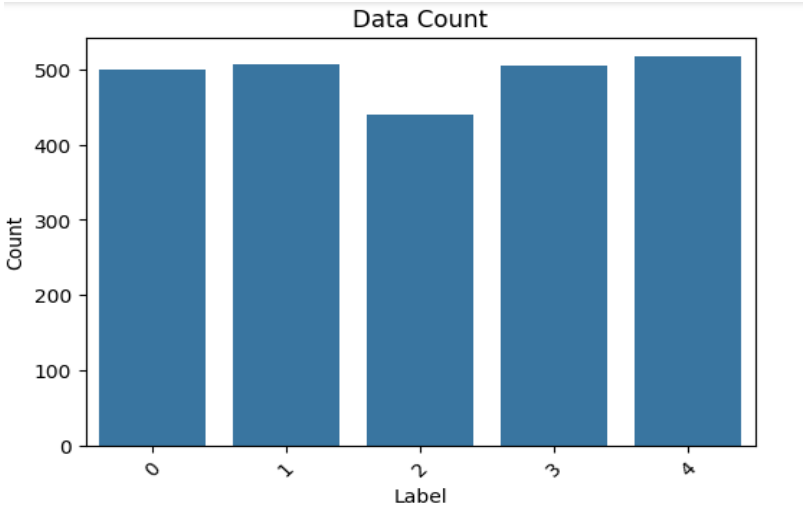


Figure 4:- Visualization of Size of each class dataset

Table 3 Source of Dataset

Class name	YouTube channel link	Amount of video
Jimma	https://www.youtube.com/watch?v=1sMVeJ8R6o4&list=PLffO18BBHSvDJb3qF94v2lgaK2o1uklzl	20
jimma	https://www.youtube.com/watch?v=qfXjAX6_vAk&list=PLy5Z-9eOjoxszda9ZfpqCSHPzx0dz7di0	22
Borena	https://www.youtube.com/watch?v=r6rL2Pckobc&list=PLZnJuyighLoyj6iwsIqx0Bo4usipuO8Lk	33
Borena	https://www.youtube.com/watch?v=VYw8j1Xgqfk&list=PLReVUPsXyPg4Yx1fut6egW_u_JtsyZSFq	27
Hararge	https://www.youtube.com/watch?v=v6KkMf5LAIg&list=PLNtqTIeXqAoFayeUvjF0wXOGlF9dYnK2u	43
Hararge	https://www.youtube.com/watch?v=ajEvTBoOb8c&list=PLNWKNOquyjwPekuup3EMN94784FlQ8oBG	10
Hararge	https://www.youtube.com/watch?v=u1J2rIQqbUo&list=PLUXJAtnOQke45iTgysJTma0BNa2mILBXw	5
Showa	https://www.youtube.com/watch?v=PH1hiyQZed0&list=PLD99610DCFF9B939A	42
showa	https://www.youtube.com/watch?v=3e3uWECRJK0&list=PLa8aDfdA3-f7WiJ6S5aoW9Q7nAf_S6izO	18
Wollega	https://www.youtube.com/watch?v=wAhQil6nKtc&list=PLwzqH-1roCuvYCrGCN0VRGbro3jRcHusW	19
Wollega	https://www.youtube.com/watch?v=bT_748DgQBA&list=PLdyZuAMsupSp2VTfjz7RKy8xnohEKFtJ1	42

Generally, for this work we have created rich collection of audio data of 2454 with the total time of 23 hours, 32 minutes, and 30 seconds (23:32:30) and 3.091GB from the five Oromo tribes. Our dataset comprises a rich collection of audio data from the five Oromo tribes, facilitating a comprehensive analysis of their unique music styles. The segmentation process allows for manageable audio clips that enhance the efficiency of our classification task.

Table 4 Summary of data used for this work

label	Original audio in number	Each audio segmented to	Length	Megabyte(Mb)
Borena	60	501	4:55:00	632
Hararge	58	503	4:28:30	633
Jimma	42	445	3:43:00	560
Showa	60	500	5:42:00	630
Wollega	61	505	4:44:00	636

3.6. Data Preprocessing

Preprocessing is a crucial step in preparing audio data for classification tasks, especially for music classification related to the Oromo tribes. In this thesis, we employed a supervised machine learning approach, which requires a labeled dataset for training. This dataset was annotated by music experts. We then divided the data into training and testing sets using the train_test_split function from the scikit-learn library. The training set was used to train the classifiers, while the testing set was utilized to evaluate their accuracy. We followed an 80/20 split, allocating 80% of the dataset for training and 20% for testing, as the size of our dataset allowed for this division. The train_test_split function offers a fast and straightforward method for this task, facilitating the analysis of testing errors. The following steps outline the preprocessing methods applied to the collected audio data.

3.6.1 Audio Format Conversion

The original audio files downloaded from video sources were primarily in MP3 format. To ensure compatibility with the machine learning models, all audio files were converted to the .wav format, which is commonly used for audio processing.

3.6.2. Segmentation

Each audio file was segmented into 30-second clips to manage computational resources effectively and facilitate analysis. This approach allows the classification models to work with manageable audio lengths while preserving the essential features of the music. For instance, the Borena tribe's audio files, originally totaling 60 entries, were segmented into approximately 590 clips, while other tribes followed similar segmentation protocols based on their respective original audio lengths.

3.6.3. Omission of Initial Segments

The first 30 seconds of each audio clip were omitted to eliminate any non-musical content, such as promotional material or studio introductions, which may not be relevant for classification. Segmentation began from index one (i+1) to ensure that the analysis focuses solely on the musical content.

3.7. Feature Extraction

Once the audio data is preprocessed and segmented, feature extraction techniques are applied to derive meaningful characteristics from the audio clips. Typically, the output from audio classification serves as the input for audio identification (Dagnew, 2023). These processes help narrow the search space, enhance processing speed, and improve result retrieval. Audio features can be categorized into temporal and spectral features, which capture the temporal and spectral characteristics of an audio signal, respectively.

In our case, for the following extracted features, the sample rate is set at 44100 Hz, a standard rate for audio processing, ensuring that the audio quality is adequate for detailed analysis. The frame size is defined as 2048 samples, which determines the number of samples per frame for analysis.

The hop size is set to 512 samples, defining the step size between consecutive frames during feature extraction. Common features used for this music classification include:

3.7.1. Mel-Frequency Cepstral Coefficients (MFCCs)

Mel-Frequency Cepstral Coefficients (MFCC) are a set of features commonly used in audio processing, speech recognition and best for recognizing structure of music signals (Kour et al., 2015). They represent the short-term power spectrum of sound, are derived from the cepstral representation of audio signals and capture the timbral aspects of the audio (Thiruvengatanadhan, 2018). MFCC is utilized to capture and represent the physiological characteristics of human perception based on the signal's frequencies, as it closely aligns with the human auditory system. It serves as a short-time power spectral representation of sound. MFCCs derive features from the cestrum of the audio signal by transforming input signals from the time domain into the frequency domain. They are widely employed in audio recognition and are increasingly being used in music information retrieval and genre classification systems.

$$C_i n = \frac{1}{T-1} \sum_{m-1}^{m} S(m) cos\left[\frac{\pi n(m=0.5)}{m}\right], 0 \leq m < m$$

where n represents the number of MFCCs, Ci(n) denotes the n-th MFCC coefficient for the i-th frame, S(m) is the logarithmic power spectrum of the audio signal, and M indicates the number of triangular filters (Xie et al., 2012).

MFCCs are widely used in audio processing and speech recognition due to their ability to represent the power spectrum of a sound signal in a way that aligns with human auditory perception. They capture the timbral and textural aspects of audio, making them particularly effective for music classification (Hossan et al., 2010).

So, since, Oromo music features distinct tonal qualities and rhythmic patterns. MFCCs can effectively capture these nuances by modeling the variations in frequency over time. This is crucial for distinguishing between different styles and genres within Oromo music, which may have specific melodic and rhythmic characteristics.

3.7.1.1 Time and Frequency in MFCC Extraction

In this research, the extraction of Mel-Frequency Cepstral Coefficients (MFCCs) is a crucial step for analyzing the characteristics of Oromo Tribe music style. The process involves both time and frequency dimensions, which significantly impact the analysis and classification of audio signals.

The audio signal is segmented into overlapping frames to capture its temporal dynamics. Each frame corresponds to a specific time window of the audio, allowing for a detailed analysis of how the music evolves over time. In our implementation, the hop length parameter is set to 512 samples, defining the step size between consecutive frames. This overlap ensures that the temporal characteristics of the music are preserved, providing a continuous representation of the audio's progression.

The other is Fourier Transform (FT) is applied during the MFCC extraction process to convert the time-domain signal into the frequency domain. This transformation enables the analysis of frequency components within each frame of the audio signal. The number of Fast Fourier Transform (n_fft) parameter is set to 2048 samples, which determines the number of points used in the Fourier Transform, thus influencing the frequency resolution of the analysis. A larger frame size offers better frequency resolution but may reduce temporal resolution.

Furthermore, mapping to the Mel scale is crucial, as it aligns the frequency representation with human auditory perception. This mapping emphasizes lower frequencies while compressing higher frequencies, making the feature set more relevant for classification tasks.

The resulting matrix of Mel-frequency cepstral coefficients (MFCCs) contains both time and frequency information, with columns representing time frames and rows corresponding to different MFCC coefficients. Each column captures the spectral properties of the audio at a specific moment, providing a compact and informative feature set. This structured representation is essential for the machine learning models used in this research, as it enables effective classification of audio segments based on the extracted features.

3.7.2. Root Mean Square (RMS)

Root Mean Square (RMS) is a statistical measure used to quantify the average power of an audio signal (Bahuleyan, 2018). It calculates the square root of the average of the squares of the audio samples, providing an indication of the signal's loudness. In the context of music classification, RMS is important as it helps differentiate between various sections of a piece based on their energy levels. RMS is a measure of the energy or power of a sound signal. It is useful for understanding the loudness and intensity of the audio, which can vary significantly between different musical performances.

Relation to Oromo Music: The dynamics of Oromo music, including variations in volume during performances, are essential for expressing emotion .RMS can help identify these dynamics, allowing for a better classification of the music based on its expressive qualities. . Higher RMS values indicate louder sounds, while lower values correspond to quieter passages.

For instance, certain Oromo music styles may exhibit stark contrasts in loudness, which RMS can capture effectively. Example showa music and Jimma music are very different in loudness. The following are some visualized RMS for two different Borena Oromo audio.

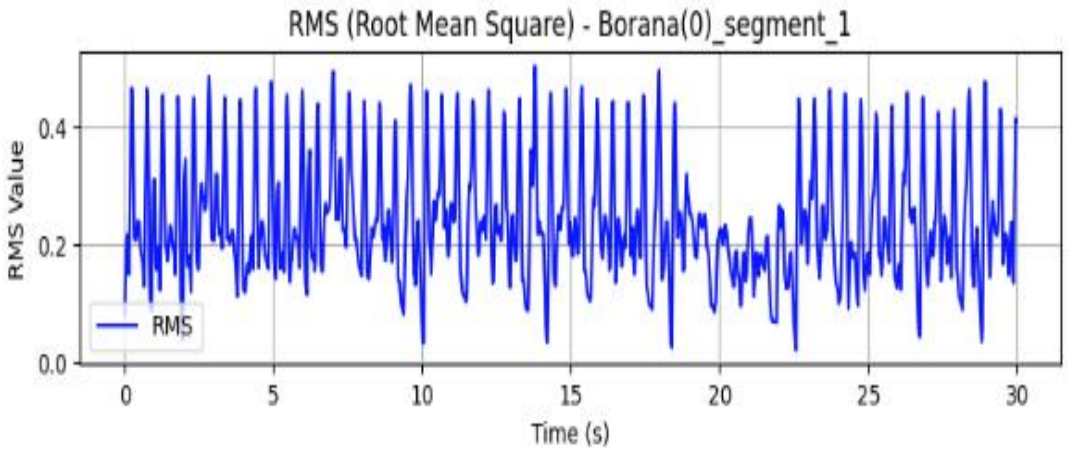
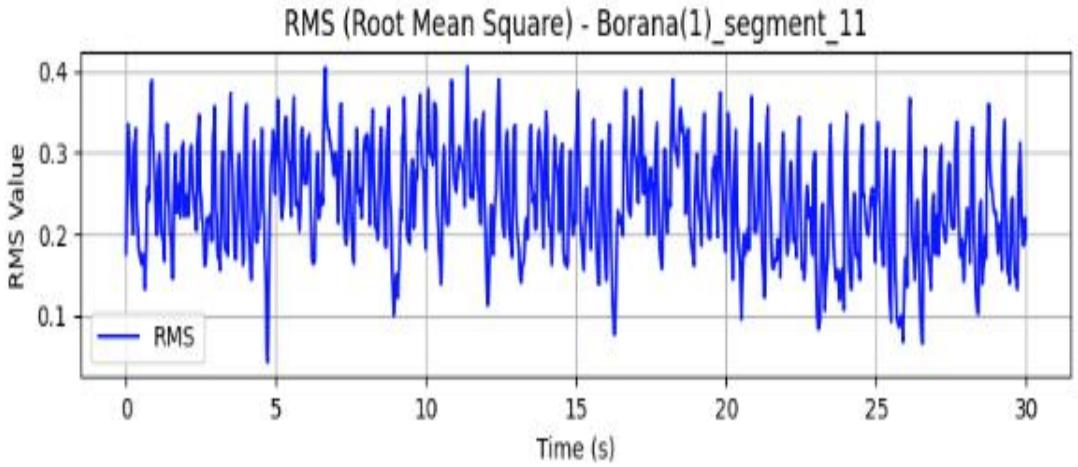


Figure 5: Visualization of RMS for audio



3.7.3 Spectral Features

Spectral features encompass a range of characteristics derived from the frequency content of an audio signal. These features can include various metrics such as spectral centroid, spectral bandwidth, and spectral flatness, each providing unique insights into the audio's tonal qualities. For instance, the spectral centroid indicates the "center of mass" of the spectrum, reflecting the brightness or darkness of the sound, while spectral bandwidth measures the spread of frequencies (Tjoa, 2017).

By analyzing these spectral features, our research captures the intricate details of the audio's tonal texture, allowing for more effective classification of musical pieces based on their harmonic and melodic content. The following are some visualized Spectral Bandwidth for two different Borena Oromo audio.

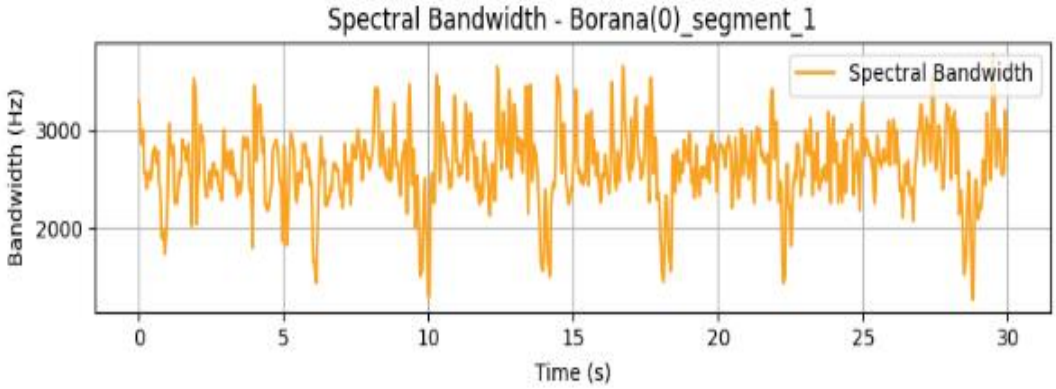
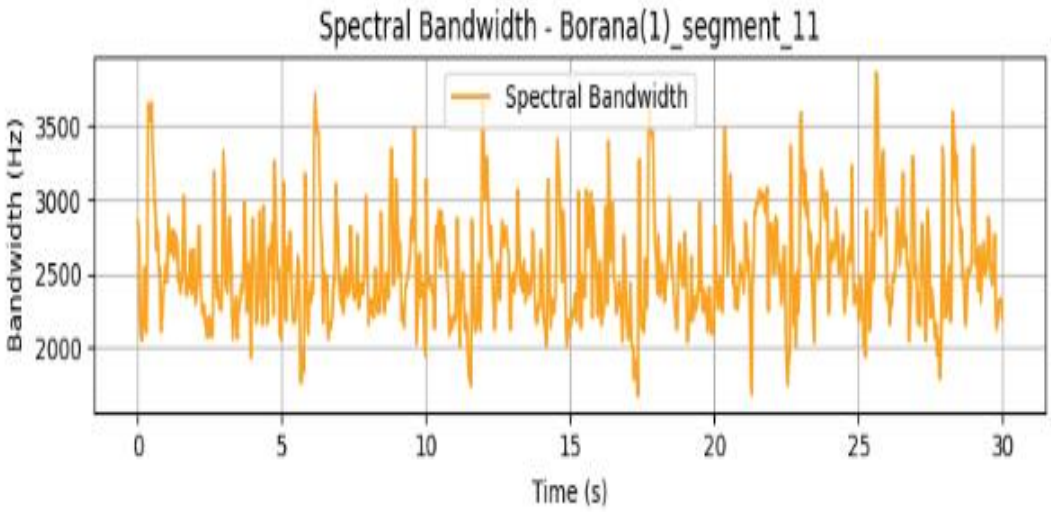


Figure 6: Visualization spectral Bandwidth



3.7.4. Rhythmic Features

3.7.4.1 Tempo

Tempo refers to the speed at which a piece of music is played, typically measured in beats per minute (BPM). The functionality in librosa first calculates a tempogram based on the method outlined by (Grosche & Muller, 2010) and then it estimates a single tempo value. It serves as a critical feature in music classification, as different genres often exhibit distinct tempo ranges.

By incorporating tempo as a feature, our research can effectively differentiate between musical styles based on their rhythmic patterns, enhancing the overall classification accuracy. Understanding tempo also provides insights into the emotional and energetic qualities of a piece, further enriching the analysis.

3.7.4.2 Average Beat Frame

Average Beat Frame is other time based feature that summarizes the timing of beats within a musical segment. It represents the average duration of beats across the audio clip, providing insights into the rhythmic structure of the music. By analyzing the Average Beat Frame, this research work identifies patterns and variations in rhythm that are characteristic of specific styles. This feature complements the tempo and enhances the overall understanding of the musical structure, contributing to more

accurate classification outcomes. Here the following is some Average beat frame visualized from two sample Hararge audio data.

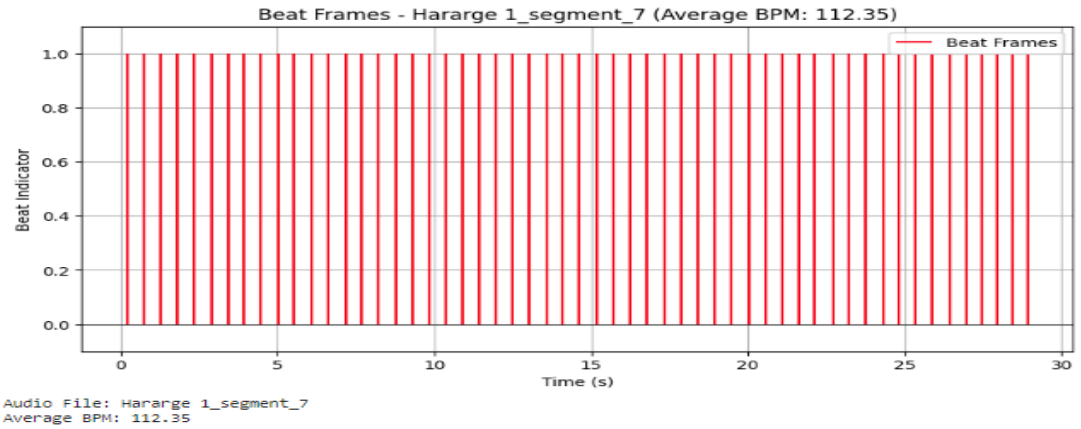
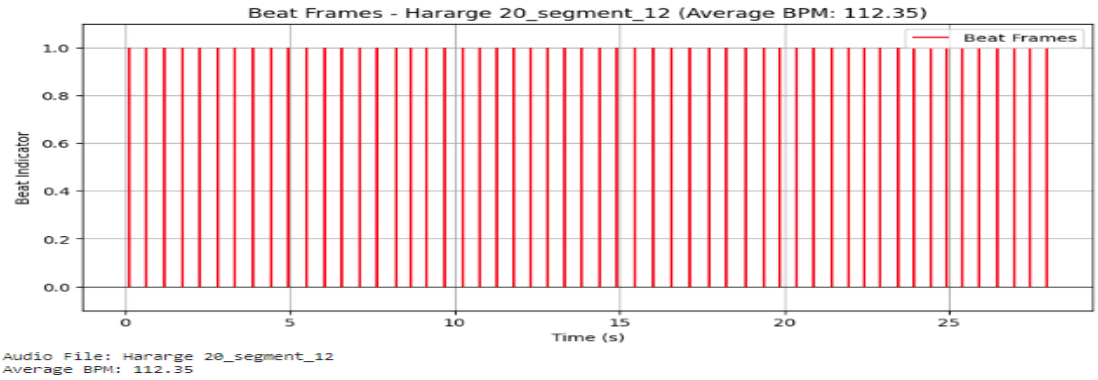


Figure 7: Visualization of Average beat frame



3.8. Algorithm selection and training

In this research, several machine learning algorithms are employed to classify Oromo music based on the extracted audio features. The selected algorithms include Random Forest (RF), Support Vector Machines (SVM), K-Nearest Neighbors (KNN. Additionally, advanced boosting techniques such as XGBoost and Gradient Boosting are utilized to enhance classification performance.

3.9. Evaluation and Performance Metrics

Evaluation metrics were employed to assess the generalization ability of the trained classifier, measuring and summarizing its performance when tested with unseen data (Huang et al., 2022). The classification models are evaluated using key performance metrics, including accuracy, precision, recall, and F1-score. This evaluation process involves assessing the models' capability to accurately classify Oromo music styles based on the unseen test data. A comparative analysis is conducted to identify the most effective algorithms and techniques for classifying Oromo music styles.

Overall, this methodology integrates data collection, preprocessing, feature extraction, algorithm selection, training, and evaluation to achieve accurate classification of Oromo tribes based on music styles using machine learning techniques. The specific algorithms, techniques, and data preprocessing steps are further adjusted and refined according to the research objectives and available resources.

3.9.1. Accuracy

Accuracy is a straightforward performance metric that assesses the overall correctness of a classification model by calculating the proportion of correctly classified instances (both true positives and true negatives) out of the total instances evaluated (Huang et al., 2022). It is expressed as a percentage and provides a general sense of how well the model performs across all classes. While accuracy is a useful measure, it can be misleading in cases of imbalanced datasets, where one class significantly outnumbers others.

$$\textbf{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN}$$

- TP (True Positives): The number of songs that are actually classified as belonging to a specific Style and are correctly identified as such.
- TN (True Negatives): The number of songs that do not belong to a specific Style and are correctly classified as not belonging to that genre.
- FP (False Positives): The number of songs that do not belong to a specific Style but are incorrectly classified as belonging to that Style and FN (False Negatives): incorrectly classified as not belonging to that Style.

3.9.2. Precision

Precision is a performance metric that measures the accuracy of positive predictions made by a classification model(Huang et al., 2022). It is defined as the ratio of true positive predictions to the total number of positive predictions (both true positives and false positives). In other words, precision indicates how many of the predicted positive instances were actually correct. This metric is particularly important in scenarios where the cost of false positives is high, as it reflects the reliability of the model in identifying relevant instances. For example, in music classification, high precision ensures that when a model predicts a specific genre, it is likely to be correct, thereby minimizing misclassifications.

- Precision = $\frac{TP}{TP + FP}$

3.9.3. Recall

Recall, also known as sensitivity or true positive rate, measures the model's ability to correctly identify all relevant positive instances within a dataset. It is defined as the ratio of true positive predictions to the total number of actual positive instances (the sum of true positives and false negatives). Recall is particularly important in situations where missing a positive instance has significant consequences, such as identifying specific musical style that may be critical for understanding cultural contexts. In music classification, high recall ensures that the model captures as many relevant pieces as possible, even if it means accepting some false positives, thus contributing to a more inclusive classification outcome.

- Recall = $\frac{TP}{TP + FN}$

3.9.4. F1-Score

The F1-score is a harmonic mean of precision and recall, providing a single metric that balances both aspects of model performance. It is particularly useful in situations where there is an uneven class distribution or when the cost of false positives and false negatives varies. The F1-score ranges from 0 to 1, with 1 being the best possible score, indicating perfect precision and recall. In music classification tasks, the F1-score is valuable because it emphasizes the importance of both correctly identifying positive cases and minimizing false negatives…… **F-Score = $\frac{2*(Precision*Recall)}{Precision + Recall}$**

CHAPTER FOUR: RESULT AND DISCUSSION

4.1 Introduction

This chapter presents the results from experiments conducted using six classifiers: RF, SVM, KNN, XGBoost, Gradient Bosting and AdaBoost. The Random Forest (RF) classifier and Support Vector Machine (SVM) were implemented using the scikit-learn library in Python. The k-Nearest Neighbors (KNN) algorithm was also built with scikit-learn. EXtreme Gradient Boosting (XGBoost) was implemented using the XGBoost library, while Gradient Boosting and AdaBoost were both developed using scikit-learn. We utilized evaluation metrics including precision, recall, accuracy, and F1 score to evaluate the classifiers' performance in this study.

4.2. Extracted Features from Audio

In the realm of music classification using machine learning, the careful selection of audio features plays a crucial role in achieving high performance. This study emphasizes the effectiveness of various spectral and temporal features, particularly those that capture the dynamics of sound. Among the features utilized are 'RMS Mean' and 'RMS Variance', which provide insights into the energy levels of the audio signals. Spectral features such as 'Spectral Centroid Mean and Var' and 'Spectral Bandwidth Mean and Var' help to characterize the distribution of frequency components, revealing essential characteristics of different musical styles. Additionally, the Mel-frequency cepstral coefficients (MFCCs), including 'MFCC1-13_Mean and Var ‘coefficient and others, are employed to extract timbral features that reflect the phonetic and tonal qualities of music.

The variance of these MFCCs further enhances the model's ability to differentiate between subtle variations in sound. The inclusion of 'Tempo' and 'Average Beat Frame' is particularly pertinent, as these temporal features encapsulate the rhythmic aspect of the music, which is vital for style classification. By leveraging these features, the classification model demonstrates improved performance, underscoring the importance of both spectral and temporal characteristics in the analysis of musical compositions.

In conclusion, the analysis demonstrates that spectral features and tempo significantly enhance the performance of music classification models. As illustrated in the accompanying graph, features such as RMS Mean, Spectral Centroid Mean, and various MFCCs consistently contribute to improved accuracy in classification tasks. The prominence of tempo further underscores its importance in distinguishing musical genres and styles. These findings highlight the effectiveness of utilizing both spectral and temporal characteristics, providing a robust foundation for future research and applications in music information retrieval. Generally Extracted and used features for this work are shown as below graph:-

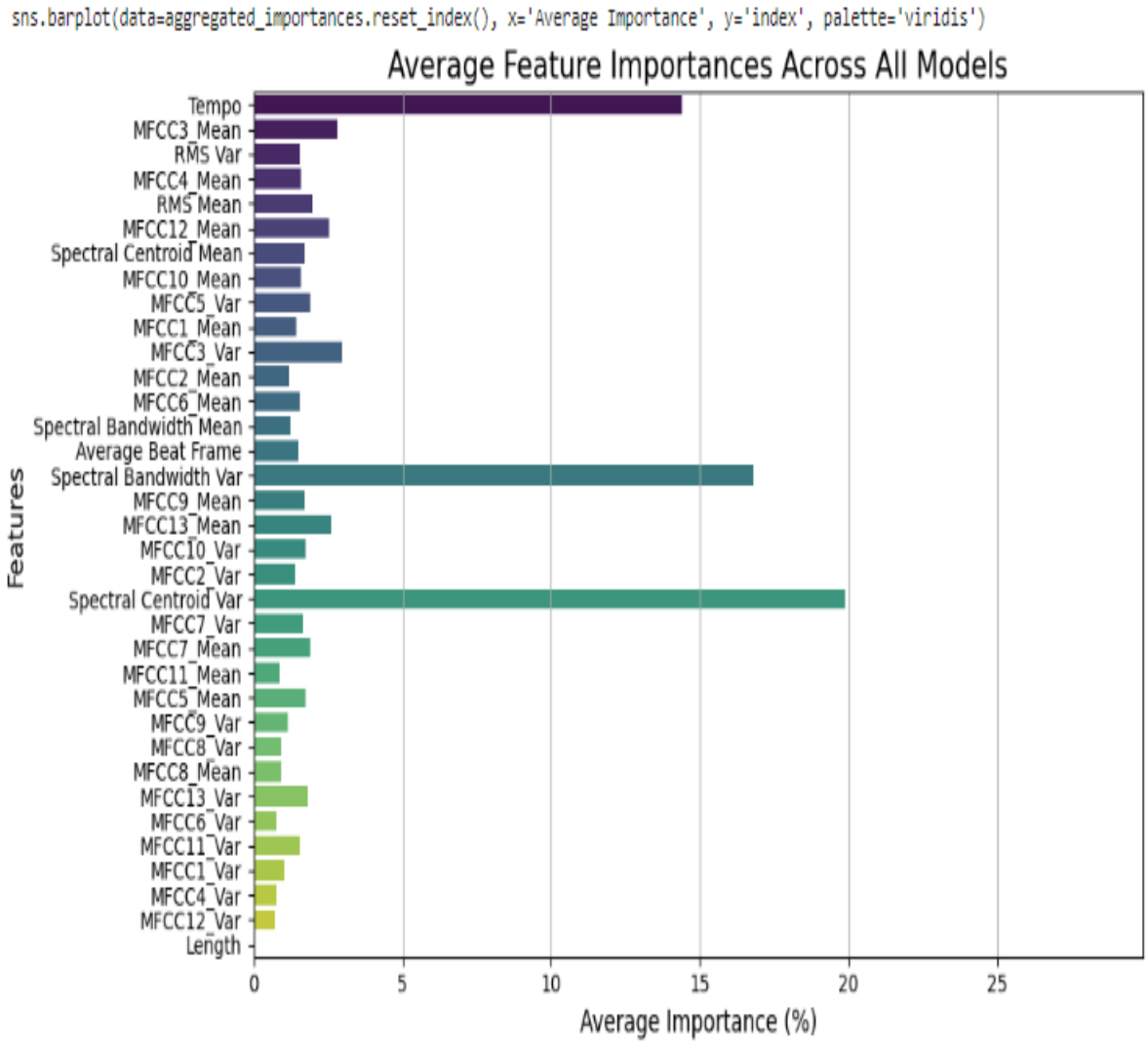


Figure 8 Feature importance

4.3. Model performance

4.3.1 Result of Random Forest

In our analysis, we implemented the Random Forest classifier to evaluate its performance on the dataset. After data preprocessed, Features explained in Section 2.2 had been extracted from audio to serve as an input for this classifiers. After conducting a hyper parameter tuning process using Grid Search, the optimal parameters identified for the model were as follows: a maximum depth of 20, log2 for the maximum features, a minimum samples leaf of 1, a minimum samples split of 2, and 200 estimators (trees in the forest). These settings were selected to balance model complexity and performance.

The performance of the Random Forest model was evaluated using accuracy, Precision, F1_score and Recall, a detailed classification report, and a confusion matrix. The best parameters identified during grid search were printed, highlighting the optimal configuration for the model. The classifier achieved an accuracy score of 92% indicating its effectiveness in distinguishing between classes. The classification report provided insights into precision, recall, and F1-score for each class, while the confusion matrix visually represented the model's performance, showing the distribution of true positives, true negatives, false positives, and false negatives as follow:-

```
Best Parameters: {'max_depth': 20, 'max_features': 'log2', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200}
Classification Report -Random Forest:
```

Classification Report -Random Forest:

	precision	recall	f1-score	support
0	0.96	0.93	0.95	103
1	0.94	0.90	0.92	97
2	0.96	0.81	0.88	86
3	0.93	0.98	0.96	109
4	0.85	0.97	0.91	99
accuracy			0.92	494
macro avg	0.93	0.92	0.92	494
weighted avg	0.93	0.92	0.92	494

A heatmap of the confusion matrix was generated using seaborn, providing a clear visual representation of the classifier's performance across different classes for this Random Forest classifier is as follow:-

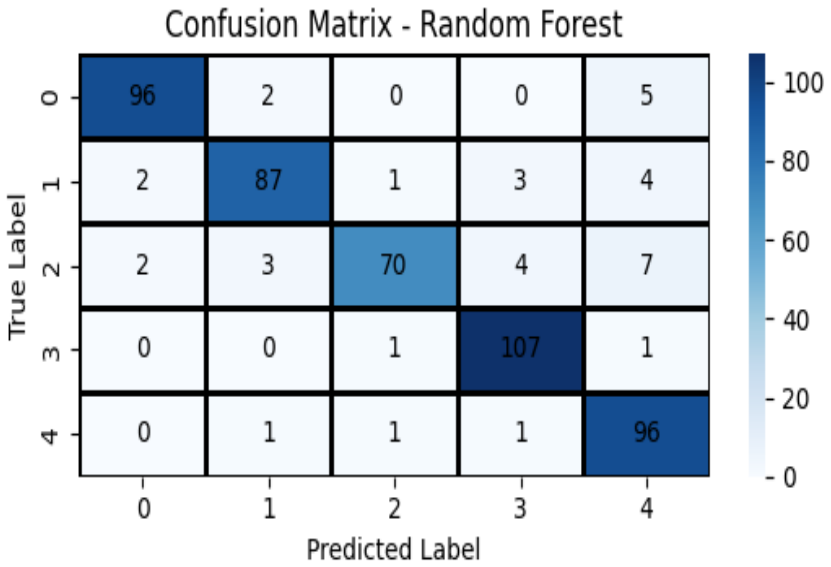


Figure 9 RF confusion matrix

4.3.2 Support Vector Machine (SVM)

The Support Vector Machine (SVM) classifier was implemented using the SVC class from the scikit-learn library. To optimize the model's performance, a grid search was conducted over several hyper parameters, including the regularization parameter CCC, kernel types (linear, RBF, and polynomial), and the gamma parameter. The grid search was configured to evaluate the model's accuracy using 5-fold cross-validation. After fitting the grid search on the training data, the best hyper parameters identified were: **C = 10**, **kernel = 'rbf'**, and **gamma = '0.1'**. The optimized model was then used to make predictions on the test dataset. The accuracy of the model was calculated, along with a detailed classification report and confusion matrix to evaluate its performance. The classification report indicated the model's precision, recall, and F1-score across different classes, while the confusion matrix provided insight into the correct and incorrect predictions.

The results reveal the effectiveness of the SVM in classifying the music data. The results are summarized as follows:

```
Fitting 5 folds for each of 36 candidates, totalling 180 fit
Best Parameters: {'C': 10, 'gamma': 0.1, 'kernel': 'rbf'}
Classification Report - SVM:
              precision    recall  f1-score   support

           0       0.96      0.99      0.98       103
           1       0.95      0.97      0.96        97
           2       0.99      0.94      0.96        86
           3       0.95      0.97      0.96       109
           4       0.98      0.95      0.96        99

    accuracy                           0.97       494
   macro avg       0.97      0.96      0.97       494
weighted avg       0.97      0.97      0.97       494
```

The confusion matrix was visualized using a heatmap, allowing for an intuitive understanding of the model's performance across different classes. Here is the visualization of CM for SVM result:-

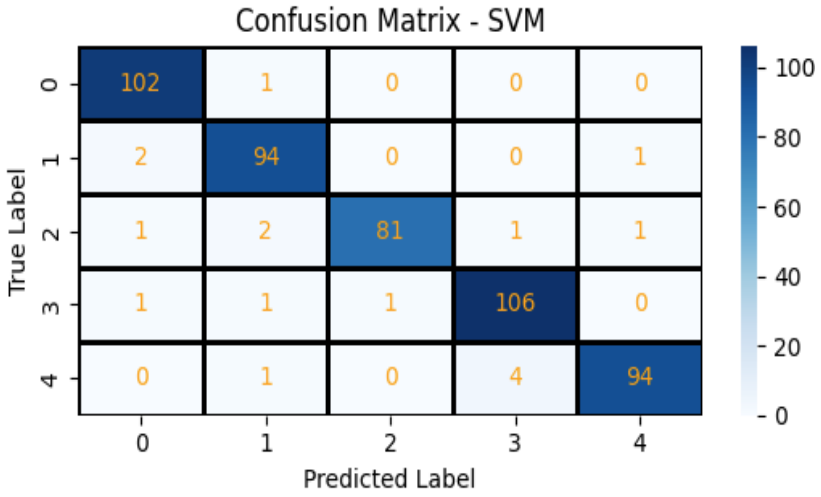


Figure 10: SVM confusion matrix

4.3.3 Result of KNN

The k-Nearest Neighbors (KNN) classifier was implemented using the scikit-learn library, and hyper parameter optimization was performed through Grid Search. The parameter grid included variations in the number of neighbors, weights, and distance metrics. The best parameters obtained from the grid search were: **n_neighbors = 3**, **weights = 'distance'**, and **metric = 'Manhattan'**. After training the model with these optimal settings, predictions were made on the test dataset.

Overall, the KNN classifier demonstrated its capability to effectively differentiate between classes based on the selected features. The summary of parameter used in this work is as follow:-

Table 5 Hyper parameter used for KNN

Hyper parameter	**values**
n_neighbors	3, 5, 7, 9, 11
Weights	uniform, distance
Distance Metric	Euclidean, Manhattan

The model achieved an accuracy score of 98%, indicating a strong performance in classifying the test samples. The classification report revealed the precision, recall, and F1-score for each class, providing a comprehensive view of the model's effectiveness. The confusion matrix illustrated the distribution of true positive, true negative, false positive, and false negative predictions, allowing for an in-depth analysis of classification errors. The overall result was shown below:-

```
Fitting 5 folds for each of 20 candidates, totalling 100 fits
Best Parameters: {'metric': 'manhattan', 'n_neighbors': 3, 'weights': 'distance'}
Classification Report - KNN:
              precision    recall  f1-score   support

           0       1.00      0.97      0.99       103
           1       0.99      0.98      0.98        97
           2       0.99      0.98      0.98        86
           3       0.96      0.99      0.98       109
           4       0.97      0.99      0.98        99

    accuracy                           0.98       494
   macro avg       0.98      0.98      0.98       494
weighted avg       0.98      0.98      0.98       494
```

The confusion matrix was visualized using a heatmap, allowing for an intuitive understanding of the model's performance across different classes. Here is the visualization of CM for KNN result:-

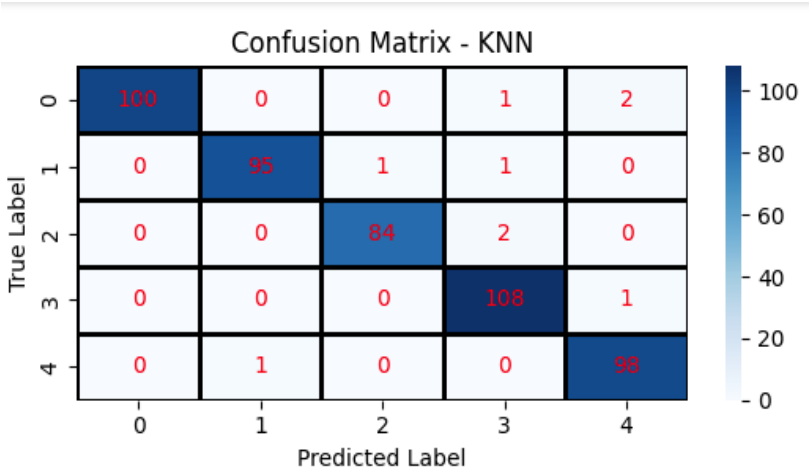


Figure 11: KNN confusion matrix

4.3.4 Result of eXtreme Gradient Boosting (XGBoost)

The eXtreme Gradient Boosting (XGBoost) classifier was implemented to enhance the predictive performance of our model. Through Grid Search, we optimized several hyper parameters to achieve the best configuration for our dataset. The model yielded an impressive accuracy of 95%, indicating its effectiveness in accurately classifying the data. The best hyper parameters identified during the search included a specified number of estimators, learning rate, maximum depth of the trees, subsample ratio, and feature sampling ratio. The classification report details the precision, recall, and F1-score for each class, while the confusion matrix visualizes the model's predictions against the actual labels. This robust performance underscores the model's ability to generalize well to unseen data.

Table 6 : Best Hyper parameters for XGBoost

Hyper parameters	Values
n_estimators	200
learning_rate	0.1
max_depth	5
subsample	0.8
colsample_bytree	0.6

```
Best Parameters: {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8}
Classification Report - XGBoost:
              precision    recall  f1-score   support

           0       0.97      0.94      0.96       103
           1       0.95      0.95      0.95        97
           2       0.96      0.87      0.91        86
           3       0.94      0.97      0.95       109
           4       0.92      0.98      0.95        99

    accuracy                           0.95       494
   macro avg       0.95      0.94      0.94       494
weighted avg       0.95      0.95      0.95       494
```

Additionally, the confusion matrix provides insights into the true positives, true negatives, false positives, and false negatives, for detailed assessment of the model's performance across different classes. Here is visualized CM for XGBoost:-

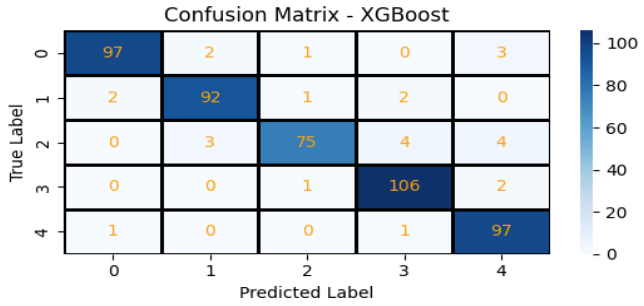


Figure 12: XGBoost confusion matrix

4.3.5. Result of Gradient Boosting

In this study, we implemented a Gradient Boosting Classifier to analyze the dataset, achieving an impressive accuracy of 96%. The model underwent hyper parameter tuning through a Randomized Search approach, optimizing several critical parameters to enhance performance. The best parameters identified during the search significantly contributed to the predictive accuracy of the model. Additionally, the confusion matrix and classification report provided insights into the model's performance, illustrating its effectiveness in correctly classifying the instances in the test dataset. Below is a summary of the best parameters found for the Gradient Boosting classifier:-

Table 7 : Best parameter for gradient boosting

Parameter	**Values**
n_estimators	100
learning_rate	0.1
max_depth	5
min_samples_split	2
min_samples_leaf	1
subsample	0.8

Generally, this classifier achieved an impressive overall accuracy of 96% across the test dataset, comprising 494 samples. The classification report reveals strong performance across all classes, with precision scores ranging from 0.94 to 0.98 and recall scores similarly high, reflecting the model's effectiveness in correctly identifying instances of each class. Notably, class 2 exhibited a slightly lower recall of 0.92, indicating some challenges in its prediction.

However, the F1-scores, which consider both precision and recall, remained robust, averaging 0.96 across all classes. These results underscore the model's reliability and effectiveness in classifying the dataset. The overall result of Gradient Boosting classifier is as follow:-

Classification Report - Gradient Boosting:

	precision	recall	f1-score	support
0	0.95	0.96	0.96	103
1	0.97	0.96	0.96	97
2	0.98	0.92	0.95	86
3	0.94	0.98	0.96	109
4	0.95	0.95	0.95	99
accuracy			0.96	494
macro avg	0.96	0.95	0.96	494
weighted avg	0.96	0.96	0.96	494

Confusion matrix provides insights into the true positives, true negatives, false positives, and false negatives, for detailed assessment of the model's performance across different classes. Here is visualized CM for Gradient Boosting:-

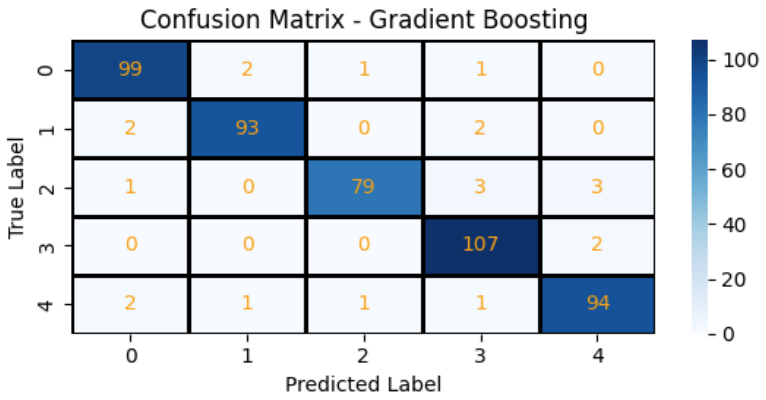


Figure 13: Gradient boosting model confusion matrix

4.3.6 Result of AdaBoost

The AdaBoost model recorded a lower accuracy on the test dataset compared with other models, indicating that it faced challenges in effectively classifying the instances. Despite employing a grid search to optimize parameters, including variations in the number of estimators and learning rates, the model's performance did not reach the levels achieved by the Gradient Boosting model.

This model achieved an accuracy of 83% on the test dataset, reflecting a notable performance gap compared to the other model. The best parameters identified through grid search included the use of a Decision Tree classifier with a maximum depth of 3, a learning rate of 1.0, and 200 estimators. The classification report reveals varying levels of precision and recall across classes, with precision scores ranging from 0.66 to 0.93. Notably, class 4 exhibited a precision of only 0.66, despite a high recall of 0.92, indicating that while the model successfully identified many true positives, it also generated a significant number of false positives.

Furthermore, the F1-scores averaged around 0.83 across all classes, suggesting that the model's balance between precision and recall could be improved. The confusion matrix illustrates these challenges, highlighting the specific classes where the model struggled most.

The following tables illustrates Parameter used during training and best parameter gained from the training of the AdaBoost model

Table 8 : Parameter used during training of AdaBoost

Parameter	Values
n_estimators	50, 100, 200
learning_rate	0.01, 0.1, 1.0
estimator	DecisionTreeClassifier(max_depth=1), DecisionTreeClassifier(max_depth=3)

Table 9 : Best parameter for AdaBoost

Parameter	Values
n_estimators	200
learning_rate	1.0
estimator	DecisionTreeClassifier(max_depth=3)

The overall report, Best parameter and confusion matrix gained from AdaBoost is shown below:-

```
Best Parameters: {'estimator': DecisionTreeClassifier(max_depth=3), 'learning_rate': 1.0, 'n_estimators': 200}
Classification Report - Adaboost:
```

```
Classification Report - Adaboost:
              precision    recall  f1-score   support

           0       0.91      0.83      0.87       103
           1       0.88      0.84      0.86        97
           2       0.85      0.70      0.76        86
           3       0.93      0.85      0.89       109
           4       0.66      0.92      0.77        99

    accuracy                           0.83       494
   macro avg       0.85      0.83      0.83       494
weighted avg       0.85      0.83      0.83       494
```

Since, confusion matrix provides valuable insights into the model's classification performance across different classes, the following CM summarizes the counts of true positive, true negative, false positive, and false negative predictions, allowing for a detailed examination of where the AdaBoost model succeeded and where it faltered.

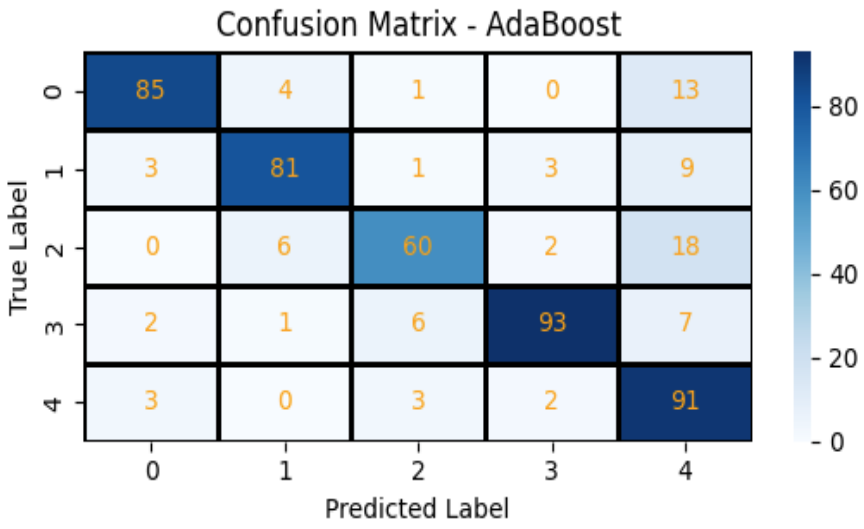


Figure 14: AdaBoost model confusion matrix

4.4 Discussion

As a result, we developed an ML based classification of five Oromo tribe music style. The feature vectors utilized include MFCC features, Tempo, spectral centroid, spectral bandwidth, Average beat frame and RMS. The system was trained using six classifiers: k-Nearest Neighbors (k-NN), support vector machines (SVM), Random Forest (RF), XGBoost, Extreme Gradient Boost and AdaBoost.

In the context of Oromo music classification using machine learning techniques, as it visualized in fig 8 above ,three key features have emerged as particularly important: tempo, Mel-frequency cepstral coefficients (MFCCs), and spectral features. Tempo serves as a fundamental characteristic, influencing the overall energy and style of the music, making it essential for distinguishing between various genres. MFCCs, which capture the spectral properties of audio signals, provide critical insights into the timbral qualities of the music, enabling the model to identify subtle variations in sound that are characteristic of different Oromo musical styles.

Our results indicated that the KNN emerged as the most effective model for this classification task, achieving the highest accuracy of **98%.** Its ability to leverage local data structures likely contributed to its success. SVM, Gradient Boosting, XGboosting and RF followed closely, demonstrating reliable performance with accuracies of 97% and 96%, 95%, 92% respectively.

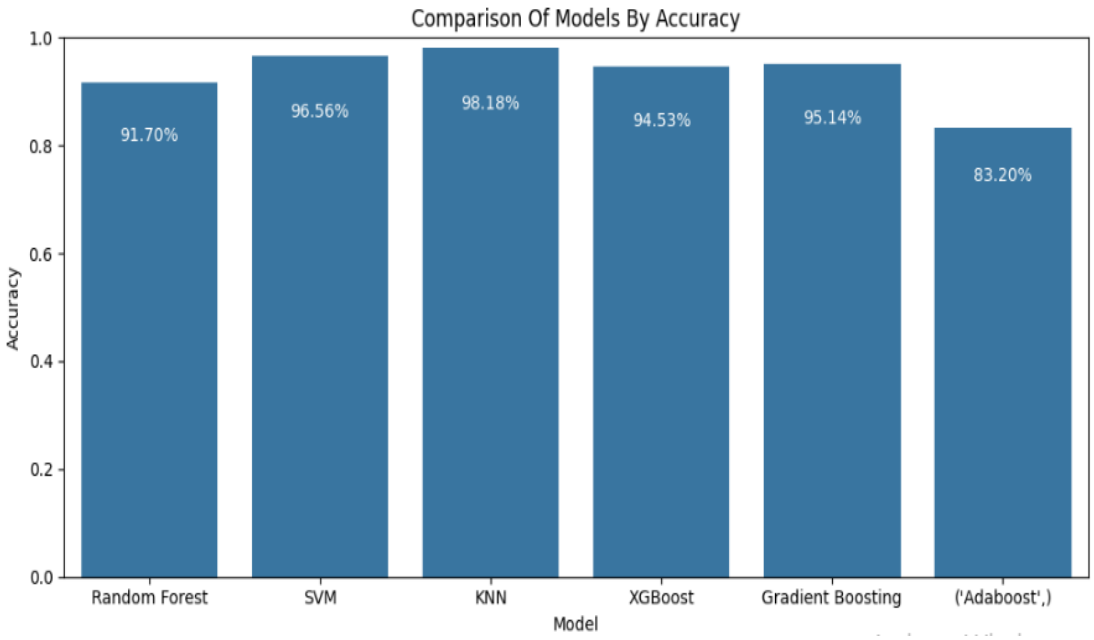


Figure 15: Overall Model Comparisons

CHAPTER FIVE

Conclusion and Recommendation

In today's digital age, the vast quantity of music released daily on the internet presents both opportunities and challenges for listeners, artists, and researchers alike. With millions of tracks spanning diverse styles and cultural backgrounds, the need for effective music classification systems has never been more critical. Efficient classification enables users to discover music that resonates with their preferences while also facilitating better organization and accessibility of content. This is particularly significant in the context of traditional music styles, such as those from the Oromo tribe, where the richness and variety of sounds can often be overlooked. By employing machine learning techniques for music classification, we can harness computational power to analyze and categorize music more accurately, thereby enhancing the listening experience and preserving cultural heritage of this region in an increasingly interconnected world.

In this research, we explored the classification of music from five distinct Oromo tribes using various machine learning techniques. Given the rich diversity of over more than five tribes, we focused on a subset to ensure adequate data availability for effective model training and evaluation. The data for this study was collected from YouTube, as detailed in Table 3.5, where a diverse range of music samples representing five distinct Oromo tribes were gathered.

This dataset provided a rich resource for exploring the classification of traditional music using machine learning techniques. Despite the limitation to five tribes due to data availability, the study successfully demonstrated the effectiveness of various classifiers in identifying and categorizing these musical styles. By extracting key features such as RMS Mean, Spectral Centroid Mean, and various MFCC metrics, we could capture the essential characteristics of the music, which played a crucial role in the classification process.

The results demonstrated that different classifiers exhibit varying levels of performance, with K-Nearest Neighbors (KNN) achieving the highest accuracy of **98.18%**. This suggests that the selected features were well-suited for distinguishing between the musical styles of the tribes studied.

The comparative analysis of the six models revealed that while KNN and Support Vector Machine (SVM) provided exceptional accuracy rates with **98.18%** and 96.56% respectively, other classifiers such as Random Forest and XGBoost also performed admirably, highlighting the potential of ensemble methods in music classification tasks.

These findings indicate that KNN and SVM are highly effective for music classification tasks, particularly within the context of Oromo tribal music. The performance of models such as Random Forest and XGBoost also highlighted their utility in this domain, while the lower accuracy of the AdaBoost model suggests that certain algorithms may be less suited for this specific classification challenge.

Despite the promising results achieved in classifying Oromo music using machine learning techniques, this research faced several limitations that should be acknowledged. Firstly, the scope of the study was limited to only five tribes within the Oromo community due to data availability issues, which may not fully represent the rich diversity of Oromo music as a whole. Additionally, the dataset was primarily sourced from YouTube, which may introduce biases related to the availability and quality of recordings. Furthermore, while several machine learning algorithms were employed, the exploration of additional models or hybrid approaches could enhance classification performance.

These limitations suggest that further research is needed to include a broader range of tribes and to utilize more diverse data sources to achieve a more comprehensive understanding of Oromo music classification. So, based on our research work we recommend future direction as follow:-

- ✓ Future studies should consider expanding the dataset to include more tribes and a broader variety of musical styles, providing a more comprehensive understanding of Oromo music and enhancing the robustness of classification models.
- ✓ Incorporating advanced preprocessing techniques and exploring deep learning approaches could further improve classification accuracy by uncovering more intricate patterns within the music data.
- ✓ Researchers are encouraged to investigate the integration of additional features, such as temporal dynamics and contextual information about the music, to enrich the dataset further.
- ✓ Collaborating with local musicians and cultural experts could provide valuable insights into the nuances of Oromo music, potentially enhancing model performance.
- ✓ This research lays the groundwork for future exploration in the classification of traditional music using machine learning, with the potential to celebrate and preserve the rich cultural heritage of the Oromo tribes.

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Appendices

Appendix A: Code for audio format conversion and segmentation using Librosa python library

+ Code + Text

```
import os
import librosa
import soundfile as sf
from pydub import AudioSegment


# Set the path to your audio files
audio_dir = '/content/drive/MyDrive/me jimmaa'

# Set the output directory for the audio files in WAV format
wav_dir = '/content/drive/MyDrive/me jimmaa/WAV FILE'

# Set the output directory for the segments
output_dir = '/content/drive/MyDrive/me jimmaa/SEGMENTED FILE'


# Loop through the audio files
for filename in sorted(os.listdir(audio_dir)):
    if filename.endswith('.webm'):
        # Load the audio file
        audio_path = os.path.join(audio_dir, filename)
        audio = AudioSegment.from_file(audio_path, format="webm")

        # Convert the audio to WAV format
        wav_filename = os.path.splitext(filename)[0] + '.wav'
        wav_path = os.path.join(wav_dir, wav_filename)
        audio.export(wav_path, format="wav")

        # Load the WAV file using librosa
        y, sr = librosa.load(wav_path)
```

```
        # Convert the audio to WAV format
        wav_filename = os.path.splitext(filename)[0] + '.wav'
        wav_path = os.path.join(wav_dir, wav_filename)
        audio.export(wav_path, format="wav")

        # Load the WAV file using librosa
        y, sr = librosa.load(wav_path)

        # Calculate the number of 30-second segments
        total_duration = librosa.get_duration(y=y, sr=sr)
        num_segments = int(total_duration // 30)

        # Loop through the segments, starting from the second one
        for i in range(1, num_segments):
            start_time = i * 30
            end_time = (i + 1) * 30
            segment = y[int(start_time * sr):int(end_time * sr)]

            # Save the segment
            output_filename = f"{os.path.splitext(wav_filename)[0]}_segment_{i}.wav"
            output_path = os.path.join(output_dir, output_filename)
            sf.write(output_path, segment, sr)
```

Appendix B: Features Used in this works

```
# Define the commonly used features based on the selection
common_features = [
    'RMS Mean', 'RMS Var', 'Spectral Centroid Mean', 'Spectral Bandwidth Mean',
    'MFCC2_Mean', 'MFCC3_Mean', 'MFCC5_Mean', 'MFCC11_Mean', 'MFCC12_Mean',
    'MFCC2_Var', 'MFCC3_Var', 'MFCC4_Var', 'MFCC5_Var', 'MFCC6_Var',
    'MFCC7_Var', 'MFCC8_Var', 'MFCC9_Var', 'MFCC10_Var', 'MFCC11_Var',
    'MFCC12_Var','Tempo','Average Beat Frame'
]
# Step 4: Feature selection (if needed)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Filter the training and test data to use only the selected features
X_train_selected = X_train[common_features]
X_test_selected = X_test[common_features]
```

Appendix C: Classifiers used

```
# machine learning models
from sklearn.ensemble import VotingClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, GradientBoostingClassifier
import xgboost as xgb

models = {
    'Random Forest': RandomForestClassifier(),
    'SVM': SVC(probability=True),
    'KNN': KNeighborsClassifier(),
    'AdaBoost': AdaBoostClassifier(),
    'Gradient Boosting': GradientBoostingClassifier(),
    'XGBoost': xgb.XGBClassifier(use_label_encoder=False, eval_metric='mlogloss')
}
```

Appendix D: Defined parameter for RF

```
# the parameter grid for Random Forest
param_grid = {
    'n_estimators': [50, 100,200],  # Number of trees in the forest
    'max_depth': [None, 10, 20,30],  # Maximum depth of the trees
    'min_samples_split': [2, 5, 10],  # Minimum number of samples required to split an internal node
    'min_samples_leaf': [1, 2, 4],  # Minimum number of samples required to be at a leaf node
    'max_features': ['auto', 'sqrt', 'log2']  # The number of features to consider when looking for the best split
}
```

Code to visualize model comparison

```
import matplotlib.pyplot as plt
import seaborn as sns

# Visualization code
metrics = list(results.keys())
accuracies = [results[model_name]['Accuracy'] for model_name in metrics]

# Bar chart visualization
plt.figure(figsize=(10, 5))
ax = sns.barplot(x=metrics, y=accuracies)
plt.ylim([0, 1])
plt.xlabel('Model')
plt.ylabel('Accuracy')
plt.title('Comparison Of Models By Accuracy')

# Add percentage labels inside the bars
for i, acc in enumerate(accuracies):
    percentage = acc * 100
    # Position the text inside the bar
    ax.text(i, acc / 1.12, f'{percentage:.2f}%', ha='center', va='center', color='white')  # Centered in the bar

plt.tight_layout()
# Hide the top spine
ax.spines['top'].set_visible(True)


plt.tight_layout()
plt.show()
```

Appendix E: Classified Oromo tribe music with each trained classifiers by taking random audio from test dataset

